Device-Aware Weighted Gerchberg-Saxton and Flicker-Bounded Phase-Continuous Sequences for Holographic Optical Trap Arrays

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Abstract

Large arrays of optical traps are computed with phase-fixed weighted Gerchberg–Saxton (WGS) on an ideal model of the spatial light modulator (SLM), and sequences that move traps interpolate between holograms never optimized for the frames between them. We test two proposed methods against phase-fixed WGS and linear phase interpolation on one scalar Fourier model. Device-aware WGS keeps the WGS weights and replaces the phase update by conjugate-gradient steps through the pixel-crosstalk kernel and the phase quantizer of the device. On a 10×10 lattice behind a crosstalk device it lowers the trap-intensity spread from a median of 5.21 percent to 0.205 percent, a geometric-mean factor of 24.4 (95 percent interval 17.5 to 34.1) over 5 seeds at 0.995 of its efficiency. The gain does not carry to every geometry: on the device of the studio's benchmark, 2 of 5 geometries meet both thresholds. An ablation shows that the device-aware phase step produces the gain; the variant without the weights, which is conjugate-gradient minimization through the crosstalk model after Schroff et al. applied to point traps, has the lower spread here and a mixed one on the benchmark lattices. Flicker-bounded sequences optimize every frame with the trap phases locked to a smooth trajectory and keep the lowest moving trap at 0.922 of its mean intensity (95 percent interval 0.913 to 0.931) against 0.644 for linear phase interpolation over 5 seeds. With frames of peak step at most 0.2 Fourier units the bound is met on 6 and 6 of 8 seeds of two long lattice moves; at equal frames on a long rearrangement, linear phase interpolation leads. Both results are simulated, not measured on hardware.

Keywords: holographic optical tweezers, weighted Gerchberg–Saxton, pixel crosstalk, phase quantization, linear phase interpolation, atom rearrangement.

1 Introduction

A phase-only SLM placed in the pupil plane of a lens forms an array of optical traps whose number now reaches the thousands. The largest published array holds about 6100 atoms in about 12,000 tweezers, and the same report states that the light field is one of the limits of the machine: the trap depth varies by 11.4 percent (standard deviation) across the sites after weighted GS optimization, the finite number of SLM pixels bounds the array, and aberrations grow with the tweezer power [6]. A small calibrated array reaches a relative standard deviation of the intensity of 0.3 percent [9], so the spread grows with the size of the array and with the distance between the model that computes the hologram and the device that displays it. Error-corrected neutral-atom processors of hundreds of physical qubits [32] and hybrid architectures that keep more than 3000 atoms for hours with a static SLM array [31] make the quality and the speed of the hologram a question of the experiment's yield, not of its elegance.

The hologram is computed, not designed, and three families of method dominate large arrays. The Gerchberg–Saxton iteration and its weighted form [1] equalize the trap intensities in the ideal model; freezing the focal-plane phase after a few iterations (phase-fixed WGS) [4] keeps the weights effective and reaches percent-level uniformity on arrays of 1500 foci; the same algorithm computes the holograms of the arrays of

about 1000 tweezers and of 11,998 sites [5, 6]. Camera feedback then corrects the residual spread on the experiment [8, 4]. The sequences that rearrange atoms with an SLM interpolate the position, the amplitude and the optical phase of every trap between two endpoint holograms and invert the transform at every step [26].

Two gaps follow. First, the solver of the large-array methods has no model of the device. The pixel crosstalk, the phase quantization and the finite fill factor of a real SLM change the trap intensities after the hologram is computed, and the experiments absorb the change with camera iterations. A model of crosstalk was placed in conjugate-gradient minimization of shaped potentials [21] and a convolution of the phase was placed in a weighted GS that, per its abstract, replaces amplitudes only [22], but no source puts a crosstalk model and a quantizer inside the weighted GS for point traps and compares it with phase-fixed WGS. Second, the intermediate holograms of a moving-trap sequence are never individually optimized; their trap intensities are uneven and the interference of the transient spots makes the intensity of a moving trap dip, down to a level that the source of the baseline links to atom loss [26].

We propose and test one method for each gap and report them as they are: with their intervals, with the baselines run by the same code on the same data, with an ablation of their components, and with the conditions under which they fail. The contributions are the following.

1. Device-aware weighted GS: the WGS weights, measured on the device's focal field, combined with conjugate-gradient steps on a device-aware loss whose gradient is closed form through the adjoint of the crosstalk operator, with a safeguard that returns the best admissible iterate (Section 4.1).

2. Flicker-bounded phase-continuous sequences: every intermediate hologram optimized with the trap phases locked to a smooth trajectory, together with a closed form for the transient intensity of a trap in the blend model and the frame count that the closed form implies (Section 4.2, Section 5).

3. A controlled comparison: both methods run against phase-fixed WGS and linear phase interpolation implemented in the same engine on the same tasks, with seeds, intervals, the benchmark geometries of the studio and long moves (Section 6, Section 7).

4. An ablation measured by a committed script, whose main outcome reverses the expected role of a component: the device-aware phase step carries the gain of device-aware WGS, and the weights lower it on the acceptance task, while on the lattices of the benchmark's device neither form dominates (Section 8).

5. The scope of each claim, stated with the geometries and moves where it fails (Section 9, Section 10).

Section 2 places the methods among the published ones, Section 3 fixes the model, the device and the metrics, Section 4 defines the two methods, Section 5 gives the propositions that explain them, and Sections 6 to 8 give the experiments. Section 11 lists the commands that rebuild every number.

2 Related work

2.1 Analytic and iterative solvers for point traps

The point-trap model of Di Leonardo et al. computes the field at each trap as a sum over SLM pixels and compares the solvers on a ten by ten lattice on a 768 by 768 pixel SLM. In simulation, plain GS reaches an efficiency of 0.94 and a uniformity of 0.60, and the weighted form reaches 0.93 and 0.99 in about 30 steps, with efficiency above 90 percent already after 10 steps [1]. Bianchi and Di Leonardo moved the random-superposition start and the weighted iteration to a GPU [2]. Direct search changes one pixel at a time: Polin et al. reached less than 5 percent deviation on 119 quasi-periodic traps where GS and the adaptive additive variant stayed above 50 percent [3], at a cost that Di Leonardo et al. measured as about three times that of GS for an eight-level phase [1]. For large arrays, Kim et al. observed that the weights lose their effect once the SLM-plane amplitude is replaced by the laser amplitude, and froze the focal-plane phases after an initial number of iterations; on 1500 foci they report a modulation efficiency of 91.2 percent and a non-uniformity below 0.5 percent in simulation [4]. The same rule computed the holograms of the arrays of about 1000 tweezers [5] and the weights of the array of 11,998 sites [6]. The open library slmsuite implements the same weighted iteration with and without camera feedback and serves as an independent check of an implementation [7]. All of these solvers optimize on an ideal model: the field they equalize is not the field the device produces.

2.2 Camera feedback

Feedback closes the gap between the model and the device by rescaling the target with the measured trap intensities. Nogrette et al. reduced the root-mean-square depth non-uniformity of ten by ten traps from 19 percent to 1.4 percent in 20 iterations with a gain of about 0.7 [8]. Kim et al. reduced a 22 percent non-uniformity to 1.59 percent in three steps with the phase-fixed rule, against 6.47 percent for conventional WGS [4]. Chew et al. report a relative standard deviation of 0.3 percent on a small array [9], and Wang et al. a non-uniformity below 1.1 percent for arrays above 1000 traps [10]; both numbers are taken from abstracts. Feedback needs the experiment and several camera exposures, and it starts from the hologram that the solver returns. A solver whose output is already flat on the device would shorten the loop; this is an expectation that the present paper does not test, since no camera is in the loop here. The two loops start from different spreads, 19 and 22 percent, and both end at about one percent, so a start that is flat on the device model would begin the loop closer to its end than a start that is flat only on the ideal model.

2.3 Cost-function, shaped-field, optimal-transport and learned holography

Methods that minimize a cost over the full field serve shaped targets. The mixed-region amplitude freedom method of Pasienski and DeMarco lowers the error on six shaped targets to 0.015 to 0.039 against 0.21 to 0.36 for GS, at an efficiency of 0.18 to 0.45 against 0.96 to 0.99 [11]. Conjugate-gradient minimization of a power of the intensity error removes the vortices that the basic cost leaves, reaching 0.07 percent after 13,000 iterations on a ring [12], and a later variant controls phase and amplitude together [13]. Optimal-transport initialization lowers the error of GS and of the mixed-region method on the same six targets, for example to 0.17 to 0.96 percent with an efficiency of 53 to 85 percent for the latter [14], and a dual formulation reduces the memory to a linear order so that a 1024 by 1024 problem runs in 76 s on a CPU [15]. Gradient methods with Wirtinger derivatives optimize any differentiable propagation model [16], and neural holography adds a calibrated model and a camera in the loop [17]. Learned generators are faster: DeepCGH is at least ten times faster than GS and 9 to 43 percent more accurate on multi-plane volumes [18], a network below 620 KB runs at video rate [19], and a network for about 20 tweezers needs 0.16 s per hologram against about 6 s for GS, with about 10 percent variation across sites [20]. None of these reports the uniformity of hundreds of point traps against phase-fixed WGS on the same array.

2.4 Device models inside the solver

The device appears in the optimization in three ways. Schroff et al. model the pixel crosstalk by upscaling each SLM pixel to three by three computational pixels and convolving the phase with a kernel fitted to a device, and minimize a normalized intensity error with conjugate gradients and automatic differentiation; the model lowered the error of a large disc by a factor of about 0.4, lowered the efficiency by about 20 percent, made the run about six times longer (about 10 minutes per hologram), and was not compared with a weighted GS [21]. Persson et al. modify a weighted GS for crosstalk, modelled as a low-pass filter of the phase retardation, and report more than 100 percent improvement of the spot uniformity on a device with large crosstalk; as far as the abstract shows, the correction replaces amplitudes only [22]. The flicker of a liquid-crystal device is a measured property: a twisted-nematic device shows phase fluctuations of about 120 degrees at one gray level [23]. Aberrations are the other device effect: a position-dependent correction extends the aberration-free field of view [24, 25]. No source found combines a crosstalk model and a quantizer with the weights of a weighted GS on point traps.

2.5 Moving traps and rearrangement

Atoms are sorted by moving traps. The hybrid architectures use acousto-optic deflectors for the fast moves of a few atoms on top of a static SLM array [29, 30, 31], and an SLM-only approach was proposed with GS, zero padding and weighted feedback [28]. The linear phase interpolation of the source of our baseline computes two endpoint WGS holograms, interpolates the position, amplitude and optical phase of each trap and inverts the transform at each step; with 13 holograms in about 40 ms it sorts a six by six array into four by four, observes that the spot of independent WGS holograms nearly vanished several times at 11 kHz sampling while the interpolated spot stayed above 70 percent, and links phase slips near π to atom loss [26]. The assignment of atoms to sites is a Hungarian matching whose exponent on the distance avoids trespassing [27]. No source optimizes the intermediate frames of a trap trajectory. The 3D extension of GS propagates plane by plane [33]; the present paper stays in two dimensions.

Table 1: Published methods that bound the two gaps: what each assumes and where it falls short for the task of the paper (uniform point traps on a device, and moving traps with bounded intensity dips). The last two rows are the proposed methods.

Work	Assumes	Falls short for the two gaps
Weighted GS [1]	ideal SLM model	equalizes the model; no crosstalk or quantizer in the loop
Phase-fixed WGS [4]	ideal model; camera feedback afterwards	spread set by the device; needs the experiment to close the gap
Direct search [3]	ideal model; one pixel at a time	cost scales with the pixel count; no device model
Mixed-region amplitude freedom [11]	shaped field; noise region	efficiency 0.18 to 0.45; not for sparse point traps
Conjugate gradient [12]	ideal model; full field	thousands of iterations; no device model
Crosstalk with conjugate gradient [21]	shaped potentials; autodiff	about 10 minutes per hologram; no weighted GS comparison
Crosstalk in weighted GS [22]	crosstalk as phase low-pass	per its abstract, replaces amplitudes only; the full text was not read, so the quantizer and the point-trap comparison are not assessed
Learned generators [20, 18]	training set per geometry class	about 10 percent variation at 20 traps; no comparison at hundreds
Linear phase interpolation [26]	two endpoint WGS holograms	intermediate frames never optimized; dips and phase slips
Device-aware WGS	device model, traps on bins	baked cost; scope measured in this paper
Flicker-bounded sequence	locked trap phases, end phases given	frames grow with the move; blend model unverified

2.6 Position of the proposed methods

Table 1 lists the works that matter for the two gaps with what each assumes and where it falls short. Device-aware WGS takes the weights of Di Leonardo et al. and the device model of Schroff et al. and joins them in a point-trap solver, which neither source does, and it is compared with phase-fixed WGS, the method that computes the holograms of the largest arrays. Flicker-bounded sequences take the phase-fixed rule of Kim et al. and the interpolation insight of the linear-phase source, and optimize each frame, which the source does not. Both are combinations of published pieces and are not claimed as new algorithms in isolation. The nearest published approach to device-aware WGS is conjugate-gradient minimization through a crosstalk model [21], and the variant of device-aware WGS without the weight update is that approach transposed to point traps on bins (our implementation of the idea, not the authors' code or loss); the ablation of Section 8 shows that this variant has the lower spread on the acceptance task, so the weights are not what the paper contributes there. What the paper contributes is the transposition and its measurement against phase-fixed WGS, the closed forms that explain the floor and the safeguard, the measurement of where it holds and fails, and, for sequences, the per-frame optimization with locked phases.

3 Problem formulation

3.1 Forward model of the traps

The SLM has $n_{\mathrm{pix}} \times n_{\mathrm{pix}}$ pixels of pitch p and a Gaussian input beam of amplitude A_j normalized so that $\sum_j A_j^2 = n_{\mathrm{pix}}^2$. A hologram is a phase $\phi_j \in [0, 2\pi)$ per pixel j at position (x_j, y_j). For a trap m at (x_m, y_m) in the focal plane of a lens of focal length f at wavelength λ, with $z_m = 0$ and no vortex, the field and the intensity are

$$V_m = \frac{1}{n_{\mathrm{pix}}^2} \sum_j A_j e^{\,i(\phi_j - \Delta_j^m)}, \qquad \Delta_j^m = \frac{2\pi}{\lambda f}(x_j x_m + y_j y_m), \qquad I_m = |V_m|^2, \tag{1}$$

following Di Leonardo et al. [1]. For traps on the bins of the zero-padded fast Fourier transform (FFT) of the SLM field, V_m is the corresponding entry of that transform. The traps are on bins throughout: the lattices are spaced a whole number of bins apart, and the focal bin is $\lambda f/(\mathrm{pad}\, n_{\mathrm{pix}}\, p)$ with a padding factor pad. The unit of length in the focal plane for moves is the Fourier unit $\lambda f/(n_{\mathrm{pix}} p)$, equal to pad bins. Each trap has a

design weight ω_m (equal for all traps in every task of the paper), a design amplitude $a_m = \sqrt{\omega_m / \langle \omega \rangle}$ and a relative amplitude $r_m = |V_m| / a_m$. Equation 1 is the model that every hologram of the paper is computed on in the ideal case and scored on when the device is the ideal one; the device of the next subsection changes what is displayed and leaves the equation as the definition of what a displayed phase produces at a trap. The equation is linear in the unit phasors $e^{i\phi_j}$, which is what makes the gradient of the intensity loss available in closed form later.

3.2 The device

A real SLM displays something other than the hologram that is computed. The displayed phase is wrapped, quantized to 2^b levels and blurred by the crosstalk between neighbouring pixels, and the pixel covers only part of its area. We use the device model of the studio [21]: each SLM pixel is $u \times u$ computational pixels, and the phase is convolved with the kernel

$$K = \mathcal{F}^{-1}\Big\{ \exp\Big[- \frac{|k_x|^q + |k_y|^q}{\sigma_K^q} \Big] \Big\}, \qquad \sigma_K = 1.24,\ q = 1.80, \tag{2}$$

the values fitted by Schroff et al. on one device (Fig. 4a). With U the replication of each pixel into $u \times u$ blocks and $\tilde{\phi} = K * (U\phi)$ the blurred phase on the fine grid, the focal field of what the device displays is

$$E = \operatorname{crop} \mathcal{F}\{ \tilde{A}\, e^{i\tilde{\phi}} \} \cdot \mathcal{E}_{\text{fill}}, \qquad V_m = E[\text{bin of } m]/\text{scale}, \tag{3}$$

where $\tilde{A}$ is the amplitude on the fine grid, the crop returns the field to the window of the ideal FFT and $\mathcal{E}_{\text{fill}}$ is the field envelope of a square pixel of fill factor fill. Two devices are used. The *crosstalk device* has the kernel above, no quantization and a fill factor of one. The *full device* adds a 4-bit quantizer and a fill factor of 0.85. The device of the studio's benchmark and of case C08 adds a random flicker of 0.05 rad rms to the displayed phase. Every hologram in the paper is scored on the device with the same function, whatever model it was optimized on, so a method is never scored on its own model unless the device and the model coincide.

3.3 Figures of merit

The efficiency is the light in the traps, and the spread is the relative standard deviation of the intensities divided by the design weights, in percent:

$$e = \sum_m I_m, \qquad \sigma = 100\, \frac{\operatorname{std}_m(I_m/\omega_m)}{\operatorname{mean}_m(I_m/\omega_m)}. \tag{4}$$

The comparison of a method with the reference solver on the same seed uses two ratios, the factor on the spread and the ratio of the efficiencies,

$$F = \frac{\min\{\sigma_{30}, \sigma_{100}\}}{\sigma_{\text{DA}}}, \qquad R_e = \frac{e_{\text{DA}}}{\max\{e_{30}, e_{100}\}}, \tag{5}$$

where the subscripts 30 and 100 name the iteration budgets of phase-fixed WGS, so the reference is the best of the two runs on the spread and on the efficiency, and the comparison is never taken against a baseline that more iterations would improve. The spread of Eq. 4 is the quantity that the experiments of the sources report as non-uniformity or as relative standard deviation, which makes the numbers of the paper comparable in kind with theirs; the uniformity defined through the maximum and the minimum intensity is not used, because it depends on two traps only. Dividing the intensity by the design weight lets the same formula serve arrays whose traps are meant to differ, although every task here has equal traps. The efficiency ratio of Eq. 5 protects the factor from a trivial gain: a hologram that sent light to the zero order would flatten the traps at a low efficiency, and the ratio would show it.

3.4 Moving traps and the transient intensity

A sequence has $S + 1$ frames $s = 0, \ldots, S$ and a trap position $x_m(s)$ that follows a time profile $h(\tau)$ with $\tau = s/S$. The SLM does not jump from one hologram to the next, and a trap that moves is read at its position between the frames. We model the displayed field between frames s and $s + 1$ as the blend of

Table 2: Notation: every symbol of the formulation with its meaning and unit. Dimensionless quantities are marked as such.

Symbol	Meaning	Unit
n_{pix}, p	SLM pixels per side, pixel pitch	1, m
λ, f	wavelength, focal length of the lens	m
A_j, ϕ_j	input amplitude and displayed phase of pixel j	1, rad
Δ_j^m	delay from pixel j to trap m	rad
V_m, $I_m = \lvert V_m\rvert^2$	trap field and intensity (fraction of the SLM flux)	1
ω_m, a_m, r_m	design weight, design amplitude, relative amplitude $\lvert V_m\rvert/a_m$	1
w_m	multiplicative weight of the iteration	1
e, σ	efficiency, relative standard deviation of the intensities	1, percent
F, R_e	spread factor and efficiency ratio against phase-fixed WGS	1
K, σ_K, q, u	crosstalk kernel, its width and exponent, upsampling factor	1
$\tilde{\phi}$	blurred phase on the fine grid	rad
L, T_m, η	device-aware loss, its target intensity, current efficiency	1
K_{cg}	conjugate-gradient steps per outer iteration	1
$x_m(s)$, $\theta_m(s)$	trap position and locked optical phase at frame s	Fourier unit, rad
$h(\tau)$	minimal-jerk time profile	1
S	number of frame intervals of a sequence	1
δ	shift of the pattern between two frames	Fourier unit
ρ_m	transient ratio of trap m	1

the two holograms, $f_\alpha = A[(1-\alpha)e^{i\phi_s} + \alpha e^{i\phi_{s+1}}]$ with $\alpha \in [0,1]$ (10 sub-frames), and read trap m at its trajectory point $x_m(s+\alpha)$. With $I_m(s,\alpha) = \lvert V_m(f_\alpha; x_m(s+\alpha))\rvert^2$ the transient ratio of a trap is

$$\rho_m = \frac{\min_{s,\alpha} I_m(s,\alpha)}{\mathrm{mean}_s\, I_m(s)}. \tag{6}$$

The blend model is our own definition: the source of the baseline gives a measurement, not an equation, so the absolute value of ρ_m is a property of the model and the methods are compared with each other only. Both sequences are scored by the same function. The numerator of Eq. 6 is the lowest intensity that trap m takes over the frames and over the blend instants, and the denominator is its mean intensity at the frames, so $\rho_m = 1$ means that the trap never dips below its mean and a value of 0.9 means that the lowest transient is a tenth below it. The blend represents the liquid crystal relaxing between two holograms; the refresh rate and the temporal response of the device are not modelled, so ρ_m is a property of the pair of holograms and of the trajectory and not of an instrument.

3.5 Notation and assumptions

Table 2 collects the symbols. The assumptions are the following. The model is scalar and monochromatic; the traps lie on FFT bins in one plane; the crosstalk parameters are those fitted on another device by the source, and their frequency unit is carried over from the model and is not verified against the source; the quantizer is differentiated as the identity (straight-through) in the optimization, and the random flicker is scored but not modelled by the optimizer; the transient model is the blend above. Each of these limits is revisited in Section 10. A further assumption concerns the baselines: phase-fixed WGS is run on the ideal model and scored on the device, which is the practice of the large-array experiments, where the solver is not given a device model and the device is corrected afterwards by camera feedback. The comparison is therefore the one that matters before feedback, and the weights, the starts and the random numbers of every method come from the same seeded generator.

4 Proposed methods

The two methods share the point-trap model of Section 3, the weighted update of Di Leonardo et al. [1] and the phase-fixed rule of Kim et al. [4], and they differ in the gap they close. Device-aware weighted GS closes the gap between the model that computes a static hologram and the device that displays it. Flicker-bounded sequences close the gap between consecutive holograms of a moving array. Both run offline: they are baked into the studio's artifacts, and the live engine of the studio does not carry them, because

the cost of one iteration exceeds the budget of an interactive frame. Each subsection gives the idea, the derivation, the algorithm, the parameters with their values and the way they were chosen, the complexity and the implementation.

4.1 Device-aware weighted GS

4.1.1 Idea

Weighted GS equalizes the trap intensities because its weights integrate the relative error of each trap: a trap that is too bright has its weight lowered at the next iteration, and the fixed point of the update is the set of equal amplitudes. On an ideal SLM the equalization is exact to the precision of the arithmetic. On a device the intensities that the weights see are those of the ideal model, so the weights equalize a field that the device does not produce. Two changes follow from this observation. First, the amplitudes r_m that drive the weights are read on the device's focal field (Eq. 3), so the integral action acts on the quantity that is scored. Second, the projection that produces the new phase from the weighted trap fields is an ideal-model operation, and no choice of weights can make the projection invert the crosstalk; we replace it by a few steps of gradient descent on a loss that is evaluated on the device and differentiated through it. The weights remain, because they supply a target that moves with the error, and the safeguard keeps the method from ever returning a hologram worse than its start.

4.1.2 Derivation

Let $V_m(\phi)$ be the trap fields of the device (Eq. 3) as a function of the SLM phase. Writing the blurred phase on the fine grid as $\tilde{\phi} = K * (U\phi)$, the field of trap m is linear in the unit phasors $e^{i\tilde{\phi}_p}$ of the fine pixels p,

$$V_m = \sum_p G_{mp}\, e^{i\tilde{\phi}_p}, \qquad G_{mp} = c_m \tilde{A}_p\, e^{-i\tilde{\Delta}_p^m}, \tag{7}$$

where $\tilde{\Delta}_p^m$ is the delay of Eq. 1 on the fine grid and c_m collects the readout scale and the pixel envelope of trap m. The loss compares the trap intensities with a target T_m that depends on the weights,

$$L(\phi) = \frac{\sum_m \big(|V_m(\phi)|^2 - T_m\big)^2}{\sum_m T_m^2}, \tag{8}$$

$$T_m = \eta\, \frac{(w_m a_m)^2}{\sum_k (w_k a_k)^2}, \qquad \eta = \sum_m |V_m|^2. \tag{9}$$

The level η is the current efficiency, so the target has the same total light as the hologram: the steps redistribute the light among the traps and do not trade efficiency for uniformity. The weights follow the rule of Di Leonardo et al., with the relative amplitudes read on the device,

$$w_m \leftarrow w_m \frac{\langle r \rangle}{r_m}, \qquad r_m = \frac{|V_m|}{a_m}. \tag{10}$$

The gradient of L is obtained in two steps. Let $N_T = \sum_m T_m^2$, $d_m = |V_m|^2 - T_m$ and $h_p = \sum_m G_{mp} d_m \overline{V_m}$. Differentiating L with respect to the blurred phase of the fine pixel p gives

$$\tilde{g}_p = \frac{\partial L}{\partial \tilde{\phi}_p} = \frac{4}{N_T}\, \mathrm{Re}\big(i\, e^{i\tilde{\phi}_p}\, h_p\big), \tag{11}$$

and the chain rule through the replication and the blur gives the gradient with respect to the SLM phase,

$$\nabla_\phi L = U^\top\big(\bar{K} * \tilde{g}\big), \qquad (U^\top z)_j = \sum_{p \in \text{block } j} z_p, \tag{12}$$

with $\bar{K}$ the reversed kernel. The matrix G never needs to be formed: h is one inverse transform of the readout of the weighted residual, and Eq. 12 is a convolution followed by a sum over each $u \times u$ block. This is the adjoint of the crosstalk operator, which the engine tests against the inner-product identity to 10^{-12}, so no automatic differentiation is needed and the gradient is exact for the smooth part of the model. The quantizer is treated as the identity in the backward pass and is applied in the forward pass (straight-through),

and the random flicker is not modelled. The minimization of L over ϕ with T fixed uses the Fletcher–Reeves conjugate-gradient recursion with a strong-Wolfe line search,

$$d_i = -g_i + \beta_i d_{i-1}, \quad \beta_i = \frac{g_i^\top g_i}{g_{i-1}^\top g_{i-1}}, \quad \phi_{i+1} = \phi_i + \alpha_i d_i, \tag{13}$$

with α_i satisfying $L(\phi_i + \alpha_i d_i) \le L(\phi_i) + c_1 \alpha_i g_i^\top d_i$ and $|g(\phi_i + \alpha_i d_i)^\top d_i| \le c_2 |g_i^\top d_i|$ for $c_1 = 10^{-4}$ and $c_2 = 0.1$; the choice $c_2 < 1/2$ keeps the Fletcher–Reeves direction a descent direction. The recursion restarts at every outer iteration, because the target of Eq. 9 changes with the weights and the conjugacy of the old directions no longer holds.

The weight update acts as an integrator and a few gradient steps may not follow it on a device that quantizes and blurs the response. The method therefore keeps every iterate ϕ^k, $k = 0, \dots, K_{\text{out}}$, with ϕ^0 the start, and returns the iterate of lowest spread on its own device model among those that keep a fraction γ of the start's efficiency,

$$k^\star = \arg\min_{k \in \mathcal{A}} \sigma\big(\phi^k\big), \qquad \mathcal{A} = \{k : \; e(\phi^k) \ge \gamma \, e(\phi^0)\} \ni 0, \tag{14}$$

with $\gamma = 0.9$, the bound of the efficiency in the hypothesis of Section 6. Because the start belongs to $\mathcal{A}$, the returned hologram is never worse than the start on the model of the method.

Three remarks read the equations. First, Eq. 8 is a loss on intensities, not on amplitudes: the intensities are what the scoring of Eq. 4 compares, and the loss does not change under a global phase of the field, so the optimization has no gauge freedom to fix. Second, the normalization by $\sum_m T_m^2$ makes the loss independent of the scale of the target, so the line search can start from a step that moves the largest pixel by a fixed fraction of a radian whatever the number of traps. Third, Eq. 9 fixes the total of the target to the current efficiency η: when a step lowers the light in the traps the next target is lower, so the loss never rewards efficiency that is lost to the zero order or to ghost orders, and the efficiency is guarded only by the floor of Eq. 14. In the ideal model, with K the identity, $u = 1$ and no quantizer, Eq. 12 reduces to the gradient of the intensity-error cost that conjugate-gradient hologram generation minimizes [12], so device-aware WGS contains that cost as its special case without a device.

The construction has limits that follow from the equations. The quantizer is a staircase: its derivative vanishes almost everywhere, so the straight-through gradient is a good direction for the smooth part of the model and carries no information about the steps of the staircase, and the full device gains less than the crosstalk device (Section 7). The readout takes the nearest FFT bin, so a trap off a bin would be read with an error, and the method refuses traps off the focal plane and vortex traps. The loss is not convex, the conjugate-gradient steps find a local descent and the result depends on the start; the start is therefore fixed to the weighted iterations from the random superposition of the seed, and the seeds of the experiments sample it.

4.1.3 Algorithm

Algorithm 1 states the procedure. The start is K_{warm} iterations of weighted GS on the ideal model from the random superposition of prisms and lenses, which is the number of weighted iterations that Kim et al. run before they freeze the phases [4]. Each outer iteration reads the trap fields on the device, updates the weights, sets the target and takes K_{cg} steps.

4.1.4 Parameters

Table 3 lists the values of both methods and where each came from. The weighted-GS warm start of 12 iterations is the number that Kim et al. use before freezing the phase. The number of conjugate-gradient steps per outer iteration, 3, is a choice of the engine: few steps keep the hologram close to the previous one and let the weights supply the integral action. The number of outer iterations, 4, is the budget of the acceptance test of the engine. The floor $\gamma = 0.9$ is the efficiency bound of the hypothesis, so that an iterate that the hypothesis would reject is never returned. The crosstalk parameters σ_K and q are the values fitted by Schroff et al. on one device, and the upsampling factor u is the one of that fit. The sensitivity to the number of steps is measured in the ablation (Section 8): one step is not enough, six steps are better than three on the crosstalk device, so the default is a compromise of cost and not an optimum. The sensitivity to the crosstalk parameters themselves is not measured.

Algorithm 1 Device-aware weighted GS. The device model $\mathcal{D}$ is the composition of replication, blur, FFT on the fine grid, crop and pixel envelope; $\mathcal{D} = \mathrm{id}$ gives the ideal model.

Input: traps $\{x_m, \omega_m\}$ on FFT bins, device $\mathcal{D}$, outer iterations K_{out}, steps K_{cg}, warm iterations K_{warm}, floor γ, seed
Output: phase $\phi^\star \in [0, 2\pi)$ per SLM pixel

$\phi^0 \leftarrow K_{\mathrm{warm}}$ iterations of weighted GS on the ideal model from the random superposition (seeded)
$w_m \leftarrow 1$; compute $V(\phi^0)$ on $\mathcal{D}$; $e_0 \leftarrow e(\phi^0)$; $\sigma_0 \leftarrow \sigma(\phi^0)$; best $\leftarrow (\phi^0, \sigma_0)$
for $k = 1, \ldots, K_{\mathrm{out}}$ **do**
    $V \leftarrow V(\phi^{k-1})$ on $\mathcal{D}$; $r_m \leftarrow |V_m|/a_m$; $w_m \leftarrow w_m \langle r \rangle / r_m$ ▷ Eq. 10
    $T_m \leftarrow \eta (w_m a_m)^2 / \sum_l (w_l a_l)^2$ with $\eta = \sum |V_m|^2$ ▷ Eq. 9
    $\phi^k \leftarrow K_{\mathrm{cg}}$ Fletcher–Reeves steps on L from ϕ^{k-1} with strong-Wolfe search ▷ Eqs. 8, 12, 13
    compute $e(\phi^k)$, $\sigma(\phi^k)$ on $\mathcal{D}$
    if $e(\phi^k) \geq \gamma e_0$ **and** $\sigma(\phi^k) < \sigma_{\mathrm{best}}$ **then** best $\leftarrow (\phi^k, \sigma(\phi^k))$
    end if
end for
return the phase of best, wrapped to $[0, 2\pi)$ ▷ Eq. 14

4.1.5 Complexity

One conjugate-gradient step evaluates the forward model on the fine grid and its adjoint once per line-search trial, about u^2 times the pixels of the plain FFT, and the weights add a pass over the traps. With $N_{\mathrm{fine}} = u^2 n_{\mathrm{pad}}^2$ the number of fine samples, a step costs $O(N_{\mathrm{fine}} \log N_{\mathrm{fine}})$ in the FFT form, and the readout at the traps is $O(M)$. An outer iteration costs K_{cg} steps plus the line-search trials, and a run costs K_{warm} ideal iterations plus K_{out} outer iterations. The pruned transform that the engine uses for the device-aware cost makes the transform about 2.4 times cheaper than the full one, and the source of the crosstalk model reports that the model alone makes a hologram about six times slower. A requirement of the research plan was that the time per iteration stay within a factor of three of the baseline; this ratio was not measured, so it is not claimed. The acceptance test of the engine takes about half a minute on the development machine, which is why the method is baked and not live. The line search dominates the cost of a step: each trial evaluates the forward model once, the strong-Wolfe search may use several trials per step, and the engine caps the evaluations of one search at 25, so the cost of an outer iteration is K_{cg} steps times a number of evaluations between one and the cap. The evaluations are counted and returned in the metadata of a run, so the cost of a hologram can be read afterwards and not only estimated.

4.1.6 Implementation

The method is the module `b1_device_wgs.py` of the engine in the repository of the studio, release 0.02.003. The device model is the module `device.py`, whose adjoint passes the inner-product identity to 10^{-12} for three devices, and the conjugate-gradient routine is shared with the other optimization methods of the studio. The traps must lie on FFT bins with $z = 0$ and no vortex when a device is given, and the method refuses a shaped-field target. The ablation script of Section 8 re-implements the loop with switchable components and checks that, with all components on, it returns the phase of the module to a difference of 0.0e+00 on the spread for every seed, so the ablated variants are the module with parts removed and not a second method. The result of a run carries the phase, a history with one entry per outer iteration (the loss, the efficiency and the spread), the trap fields on the device and the metadata of the run: the number of steps, the warm iterations, the evaluations, the final weights, the iteration that was returned, the floor and the spread and efficiency of the start. The history keeps every iterate so that a run that winds up can be inspected, and the convergence curve of Fig. 4b is read from it. The routine accepts an initial phase that replaces the warm start and raises an error for fewer than one step or a negative number of warm iterations. The random superposition of the start is drawn from the portable generator of the engine, so the result is a function of the seed.

Table 3: Parameters of the two methods and of the tasks, with their values as the engine uses them, and the origin of each value. The device values are those of the crosstalk device and the full device; the flicker belongs to the benchmark's device.

Parameter	Value	Origin
SLM pixels, padding	256^2, 2	acceptance tasks of the engine
Wavelength, focal length, pitch, NA	850 nm, 250 mm, 12.5 μm, 0.5	acceptance tasks of the engine
Crosstalk σ_K, q, upsampling u	1.24, 1.80, 3	fit of Schroff et al. [21]
Quantizer, fill factor (full device)	4 bits, 0.85	parameters of the studio's device cases, no published fit
Flicker (benchmark device)	0.05 rad rms	parameter of the studio's device case, no published fit
Device-aware WGS warm iterations K_{warm}	12	number of weighted iterations before the phase freeze of Kim et al.
Device-aware WGS steps K_{cg} per outer iteration	3	engine choice; sensitivity in Section 8
Device-aware WGS outer iterations K_{out}	4	budget of the acceptance test
Device-aware WGS efficiency floor γ	0.9	efficiency bound of the hypothesis
Flicker-bounded frames, iterations per frame	13, 10	sort of a six by six into a four by four array in the source; engine choice
Flicker-bounded cold start iterations	30	budget of the first frame
Flicker-bounded phase step bound per frame	0.3 rad	slips of several tenths of π barely mattered in the source; engine choice
Flicker-bounded step bound of the claim	0.2 Fourier units	frame count at which the lowest moving trap of the rearrangement case first reached 0.9
Transient sub-frames per interval	10	engine choice

4.2 Flicker-bounded phase-continuous sequences

4.2.1 Idea

Linear phase interpolation builds the intermediate holograms of a move from two endpoint holograms and never optimizes them: a frame is the inverse transform of interpolated spots, so its trap intensities are uneven and the transient spots interfere. A weighted GS that is run independently at every frame equalizes each frame but leaves the optical phase of every trap free, so the phases may jump between frames and the blend of two frames dips, which is the flicker that the source of the baseline observed. The proposed sequence combines the two remedies: it optimizes every frame, which equalizes the trap intensities, and it locks the optical phase of every trap to a smooth trajectory, which removes the phase jumps. With both, the target field changes slowly from one frame to the next, so the holograms change slowly and so does the blend. The bound on the phase step per frame refuses a trajectory whose phases would have to slip too fast.

The relation to the two parents is as follows. Linear phase interpolation has smooth phases and unoptimized frames; weighted GS at every frame has optimized frames and free phases. In the engine's earlier measurement on a nine-trap sort, weighted GS at every step equalized each frame and showed no flicker (a median transient ratio of 0.96 in that model), unlike the 11 kHz camera observation of the source; that measurement is recorded in the design record and is not repeated here, and we do not use weighted GS at every frame as a baseline for that reason, comparing instead with the baseline that the source of the problem uses. The sequence we propose is not a claim that free phases always flicker; it is a construction whose phase path is controlled by design, with a measurement of what that control is worth in an ablation (Section 8).

4.2.2 Derivation

Positions follow the minimal-jerk profile, the polynomial that has zero velocity and acceleration at both ends,

$$x_m(s) = x_m(0) + \big(x_m(S) - x_m(0)\big)\, h(\tau), \qquad h(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \qquad \tau = \frac{s}{S}. \tag{15}$$

The optical phases of the traps follow the same profile between the phases $\theta_m(0)$ of the first frame and end phases θ_m^{end} that are given (by default the phases are held, $\theta^{\mathrm{end}} = \theta(0)$),

$$\theta_m(s) = \theta_m(0) + \mathrm{wrap}\left(\theta_m^{\mathrm{end}} - \theta_m(0)\right) h(\tau). \tag{16}$$

Every frame $s \geq 1$ is a phase-fixed weighted GS step with the trap phases locked to Eq. 16: with the trap set $x_m(s)$ and the coefficients $c_m = w_m a_m e^{i\theta_m(s)}$,

$$\phi_j^{(s)} = \arg \sum_m c_m \, e^{+i\Delta_j^m(x_m(s))}, \qquad w_m \leftarrow w_m \frac{\langle r \rangle}{r_m}, \tag{17}$$

started from the hologram and the weights of frame $s-1$ and iterated K_{f} times. Frame 0 is a cold phase-fixed WGS of K_0 iterations whose trap phases give $\theta_m(0)$. The hard lock is the limit $\lambda \to \infty$ of a penalty $\lambda \sum_m (\arg V_m - \theta_m)^2$ added to the amplitude error, which the engine does not form. The sequence is refused when the phase step per frame, the largest change of $\theta_m(s)$ between consecutive frames, exceeds a bound $\dot{\theta}_{\max}$,

$$\max_{m,s} \left|\theta_m(s+1) - \theta_m(s)\right| \leq \dot{\theta}_{\max}, \tag{18}$$

and the bound has a direct meaning for the target of the optimization: the coefficients of two consecutive frames differ by

$$\left|c_m(s+1) - c_m(s)\right| \leq w_m a_m \left|\theta_m(s+1) - \theta_m(s)\right| + a_m \left|w_m(s+1) - w_m(s)\right|, \tag{19}$$

so a bounded phase step and weights that change little make the target field, and therefore the hologram that inverts it, change slowly from frame to frame. The number of frames that a move needs follows from the same profile. The velocity of h is $h'(\tau) = 30\tau^2(1-\tau)^2$, maximal at $\tau = \frac{1}{2}$ where it equals $15/8$, so the largest shift between two consecutive frames of a trap that travels L Fourier units in S intervals is, to first order in $1/S$,

$$\delta_{\max} = \frac{15}{8}\frac{L}{S}, \qquad S \geq \frac{15}{8}\frac{L}{\delta_{\mathrm{claim}}}, \tag{20}$$

with $\delta_{\mathrm{claim}} = 0.2$ Fourier units the step bound of the claim. The engine returns the smallest number of frames that satisfies the second relation and is at least the default number of frames F_0,

$$F(L) = \max\left\{F_0, \; \left\lceil \frac{15}{8}\frac{L}{\delta_{\mathrm{claim}}} \right\rceil + 1\right\}. \tag{21}$$

The origin of δ_{claim} is the closed form of Section 5 and a calibration on the rearrangement case, reported with its limits in Section 10.

The equations read as follows. Equation 15 fixes where each trap is at each frame, and Eq. 16 fixes the optical phase that it must have there. Equation 17 is the usual weighted back-projection with one difference: the phase of each coefficient comes from the prescribed path and not from the current field, and this is the difference that gives the method its name. The weights and the hologram are carried from frame to frame, so every frame is a few iterations away from the previous one and not a new solution: with held phases and small steps, the problem of frame s is a small perturbation of the problem of frame $s-1$. The weights integrate the amplitude error across frames as well as across iterations, which is the integral action of device-aware WGS again. In the penalty form the lock would be a soft constraint, and the optimizer could trade phase error for amplitude error; the hard lock refuses the trade. When end phases are given, Eq. 16 follows the wrapped difference, which is the short path on the circle and can differ from the path that the hologram at the destination would take on its own: the end phases are an input of the method and not a result of it. Between frames the displayed field is the blend of two holograms; the lock bounds the phase step, the optimization equalizes the frames, and the blend is then close to a hologram of an intermediate trap set.

4.2.3 Algorithm

Algorithm 2 states the procedure. The transient ratios of Eq. 6 are computed after the last frame on the blend model, and they are the figure of merit of the sequence; they are not used inside the optimization.

4.2.4 Parameters

The number of frames, 13, is the number of holograms that the source of the baseline uses to sort a six by six array into a four by four array. The number of iterations per frame, 10, is an engine choice: the engine's

Algorithm 2 Flicker-bounded sequence for a move from traps $X(0)$ to $X(S)$ with end phases θ^{end} (held by default).

Input: traps $X(0)$, destination $X(S)$ with the same ids, frames $S+1$, iterations K_0 and K_{f}, bound $\dot{\theta}_{\max}$, seed
Output: holograms $\phi^{(0)}, \ldots, \phi^{(S)}$ and transient ratios ρ_m
$\phi^{(0)} \leftarrow$ phase-fixed WGS of K_0 iterations on $X(0)$; $\theta_m(0) \leftarrow \arg V_m(\phi^{(0)})$
$\theta_m(s) \leftarrow$ Eq. 16; **if** Eq. 18 fails **then** refuse the sequence
$w_m \leftarrow 1$; $\phi \leftarrow \phi^{(0)}$
for $s = 1, \ldots, S$ **do**
    place the traps at $x_m(s)$ (Eq. 15); $c_m \leftarrow w_m a_m e^{i\theta_m(s)}$
    for $t = 1, \ldots, K_{\mathrm{f}}$ **do**
        $V_m \leftarrow V_m(\phi)$ at $x_m(s)$; $w_m \leftarrow w_m \langle r \rangle / r_m$; $\phi \leftarrow \arg \sum_m w_m a_m e^{i\theta_m(s)} e^{+i\Delta^m}$ ▷ Eq. 17
    end for
    $\phi^{(s)} \leftarrow \phi \bmod 2\pi$
end for
compute ρ_m from the blend of consecutive frames (Eq. 6); **return** holograms and ρ

acceptance test fails with no iteration, and the ablation of Section 8 shows that the lowest transient ratio is already above the bound with three iterations and below it with one. The bound on the phase step per frame, 0.3 rad, is an engine choice taken from the observation in the source that phase slips of several tenths of π barely mattered for atom survival while slips near π lost atoms; it is a bound on the phase path and not a claim about atoms. The step bound of the claim, 0.2 Fourier units, is derived in Section 5. The 10 sub-frames per interval of the transient model are a numerical resolution of the blend. Two parameters interact. With held phases the phase step is zero and the bound never binds, whereas with end phases the largest step is $\frac{15}{8}|\Delta\theta|/S$ for a phase difference $\Delta\theta$ (the same profile as Proposition 6), which is why the experiments with end phases on long moves lift the bound from 0.3 to 1 rad and why the default bound refuses a long move at the default number of frames. The state of the weights is carried across frames and not reset, because resetting it would discard the integral action. The freeze iteration of the phase-fixed rule at frame 0 is the smaller of the freeze iteration of Kim et al. and the cold-start budget.

4.2.5 Complexity

A sequence is S warm-started runs of K_{f} iterations, each iteration being one forward projection and one back-projection of the point-trap form, $O(N_{\mathrm{pix}}M)$ for M traps and N_{pix} SLM pixels, so the cost of the sequence is $O(SK_{\mathrm{f}}N_{\mathrm{pix}}M)$ after the cold start of frame 0. The frame count grows linearly with the length of the move (Eq. 20), and therefore so does the cost; this is the price of the bound and one of the limits of the method. The scoring of the transient ratio costs 10 projections per interval. The source of the baseline reports a computation of under 200 microseconds per hologram on a GPU for its own method; we do not rely on that figure, and our runs are offline. The sequence also stores every hologram, so the memory of a run grows with the number of frames as well, and the transient scoring evaluates the blend at every sub-frame of every interval; for the longest move of the paper, with 137 frames, the scoring is a larger share of the run than for the acceptance move.

4.2.6 Implementation

The method is the module `b2_flicker_bounded.py` of the engine, which provides the profile, the trajectory, the transient scoring and the frame-count rule, and which takes the destination, the number of frames, the iterations per frame, the end phases and the bound as arguments. It works in the point-trap form, so the traps need not lie on bins, in two or three dimensions, on the ideal device. The phase-fixed rule of the first frame and the weighted iteration are those of the ladder methods of the studio, so a correction of one reaches the other at once. The ablation script re-implements the sequence with switchable components and checks that with all components on it reproduces the lowest transient ratio of the module to a difference of 0.0e+00 for every seed. The call returns the last hologram as the phase, a history with one entry per frame (efficiency, uniformity and spread), the trap fields of the last frame and, in the metadata, every frame, the transient ratio per trap, the mask of the moving traps (those with a displacement), the phases of the first frame, the iterations per frame and the number of frames. It raises an error for a shaped target, for a destination that does not hold the same trap identifiers in the same order, for end phases whose number

differs from the number of traps, and for a phase step above the bound. The experiments read the lowest value over the moving traps, which is the quantity of the hypothesis, and score the traps that hold still through the per-frame spread.

5 Theory

The propositions below explain why each method should work and under which conditions; where a statement could not be proved we state it as a conjecture with the evidence for it. They are used in the experiments: Proposition 1 predicts the floor that the baseline meets on the device, Proposition 2 and Proposition 4 define what the weights and the safeguard guarantee, and Proposition 5 gives the closed form against which the measured transient ratios are read.

5.1 Why an ideal-model solver leaves a floor on the device

Proposition 1. *Let ϕ be a hologram whose ideal-model trap fields are exactly equal, $|V_m^0| = a_m$, and let the device display the phase $U\phi + \delta$ on the fine grid, with $\delta = K * (U\phi) - U\phi$ the residual of the blur. To first order in δ the relative intensity error of trap m on the device is*

$$\varepsilon_m = \frac{I_m - I_m^0}{I_m^0} = 2\,\mathrm{Re}\,\frac{\Delta V_m}{V_m^0}, \qquad \Delta V_m = i\sum_p G_{mp}\, e^{iU\phi_p}\,\delta_p, \tag{22}$$

and $\varepsilon_m = 0$ for all traps if and only if $\mathrm{Re}(\Delta V_m / V_m^0) = 0$ for all m, that is, if the blur leaves the trap intensities unchanged to first order, in particular if K is the identity.

Proof. With the linear form of Eq. 7, $V_m = \sum_p G_{mp} e^{i(U\phi_p + \delta_p)} = \sum_p G_{mp} e^{iU\phi_p}(1 + i\delta_p + O(\delta^2)) = V_m^0 + \Delta V_m + O(\delta^2)$. Then $I_m = |V_m^0|^2 + 2\,\mathrm{Re}(\overline{V_m^0}\Delta V_m) + O(\delta^2)$, and dividing by I_m^0 gives Eq. 22. The error vanishes for every trap exactly when $\mathrm{Re}(\Delta V_m / V_m^0) = 0$ for all m, which holds for $\delta = 0$. The expansion treats δ as small, and a wrapped phase violates that at its jumps between 0 and 2π, where the blur is of the order of the jump; Eq. 22 is therefore a local statement about where the floor comes from, and the claims of the paper rest on the measured spreads and not on the first-order value. □

The proposition has two consequences. First, an ideal-model solver cannot remove ε_m: the weights of weighted GS are computed from $|V_m^0|$, which is already equal, so the iteration has converged on the ideal model and has no information about ε_m. More iterations improve the ideal-model spread and leave ε_m unchanged, which is the control that the baseline at 100 iterations provides. Second, the size of δ is set by the spectrum of the hologram: in the Fourier domain the blur multiplies the phase spectrum by the transfer function of Eq. 2, $\hat{\delta}(k) = (\hat{K}(k) - 1)\widehat{U\phi}(k)$, which is small where the hologram is smooth and large where it carries pixel-scale structure. A lattice of traps far from the zero order needs a hologram with such structure, so the floor is expected to depend on the geometry of the array; the measured spreads of the baseline on the lattice and on the other geometries are read against this expectation in Section 7.

5.2 What the weights, the steps and the safeguard guarantee

Proposition 2. *The weight update of Eq. 10 is an integrator of the logarithmic relative amplitude: $\log w_m^{k+1} = \log w_m^k - (\log r_m^k - \log\langle r^k\rangle)$. A fixed point $w^\star$ of the update satisfies $r_m(w^\star) = \langle r\rangle$ for every trap, that is, equal relative amplitudes on the device, and conversely.*

Proof. Taking logarithms of Eq. 10 gives the first statement. The update leaves w_m unchanged if and only if $\langle r\rangle / r_m = 1$, which is $r_m = \langle r\rangle$ for every m; equal relative amplitudes mean $\sigma = 0$ for equal design weights. □

The proposition says what the weights can do and what they cannot: they drive the device to equal amplitudes if the iteration converges, and they act with the memory of an integrator, which is the reason why they are prone to overshoot when the plant that they control, a hologram that moves by a few gradient steps per iteration, does not follow. The next two statements make this precise.

Proposition 3. *Within one outer iteration, with T fixed, every accepted conjugate-gradient step lowers L, so $L(\phi^k) \le L(\phi^{k-1}; T^k)$. Across outer iterations nothing bounds L: the target T^{k+1} differs from T^k, and $L(\phi^k; T^{k+1})$ may exceed $L(\phi^k; T^k)$.*

Proof. The strong-Wolfe line search accepts a step only under the sufficient-decrease condition of Eq. 13, so L decreases along the steps for a fixed T. The target is a function of the weights and of η, both of which are updated between outer iterations; the loss with the new target is a different function. □

Proposition 4. *The iterate returned by Eq. 14 satisfies $\sigma(\phi^{k^\star}) \le \sigma(\phi^0)$ and $e(\phi^{k^\star}) \ge \gamma\, e(\phi^0)$, both measured on the model of the method. If the model of the method is the scoring device, as for the crosstalk device and the full device, the same holds on the score.*

Proof. The start belongs to $\mathcal{A}$ by definition, so the set is not empty and the minimum is over a set that contains the start. The minimizer has a spread no larger than the start's and, being in $\mathcal{A}$, an efficiency of at least $\gamma e(\phi^0)$. When the scoring device differs from the model, for example by random flicker that is scored and not modelled, the statement holds on the model and not on the score. □

Conjecture 1. *When the conjugate-gradient steps per outer iteration cannot follow the change of the target, the weights wind up and the loss rises across outer iterations; fixing the weights, or giving the steps more freedom, removes the windup.*

The evidence for the conjecture is of two kinds. A run on the square lattice of the benchmark on the device with flicker, recorded in the design record of the method, ran away on one seed in two inside the method's own model: the loss rose from 0.044 to 0.53 between the second and the fourth outer iteration, the weights spread to a ratio of about 1000 and the efficiency fell from 0.50 to 0.03; that run is the reason of the safeguard and is not re-run here. The ablation of Section 8 measures the second kind: with the weights fixed to one, or with six steps per outer iteration, the spread on the crosstalk device is lower than with the default. The conjecture is not proved, and the conditions under which the weights help are not characterized.

5.3 Transient intensity of a shifted pattern

Proposition 5. *Consider a single trap read in the blend model of Section 3, with a flat input amplitude over n pixels along one axis, and two consecutive holograms $\phi^{(s)}$ and $\phi^{(s+1)} = \phi^{(s)} + 2\pi\delta\, j/n$ that display the same pattern shifted by δ Fourier units, with $\phi_j^{(s)} - \Delta_j^{(s)}$ constant over the pixels (a hologram of one trap in phase at all pixels). At the half-way instant $\alpha = \frac{1}{2}$, read at the position $\delta/2$ from the trap of frame s, the intensity as a fraction of the frame's own intensity is*

$$\frac{I(s, \frac{1}{2})}{I(s)} = \operatorname{sinc}^2\Big(\frac{\delta}{2}\Big), \qquad \operatorname{sinc} x = \frac{\sin \pi x}{\pi x}, \tag{23}$$

in the limit of many pixels, and it equals the value $\rho_{0.9} = 0.9$ at the shift $\delta_{0.9} = 0.356$ Fourier units.

Proof. Let j run over n pixels centred on zero. The blend field is $\frac{1}{2} e^{i\phi_j^{(s)}}(1 + e^{2\pi i \delta j/n})$. The read-out at the shifted position adds the delay $\Delta_j^{(s)} + \pi\delta j/n$ to the trap's own delay, so the amplitude is $\frac{1}{n}\sum_j \frac{1}{2} e^{i(\phi_j^{(s)} - \Delta_j^{(s)})}\big(e^{-i\pi\delta j/n} + e^{+i\pi\delta j/n}\big) = \frac{1}{n}\sum_j e^{i(\phi_j^{(s)} - \Delta_j^{(s)})} \cos(\pi\delta j/n)$. With the phase constant the sum is $\frac{1}{n}\sum_j \cos(\pi\delta j/n) \to \int_{-1/2}^{1/2} \cos(\pi\delta u)\, du = \operatorname{sinc}(\delta/2)$, and the intensity is its square. The value $\delta_{0.9}$ is the root of $\operatorname{sinc}^2(\delta/2) = 0.9$, which the measurement script solves by bisection. □

The closed form is a reference for a single trap. In a multi-trap hologram the other traps change between frames as well, and the lowest moving trap can lie below the single-trap value. Two comparisons show how the measurements stand against it: at the peak step of the acceptance task, 0.31 Fourier units, Eq. 23 gives 0.922, and the measured lowest transient ratio is 0.922 on average; at the step bound of the claim, 0.2 Fourier units, it gives 0.968, and the lattice moves of Table 7, at peak steps of 0.197 and 0.199 Fourier units, measure medians of 0.933 and 0.936. The bound of the claim is therefore not the shift at which the single-trap form reaches 0.9, which is 0.356 Fourier units, but a more conservative step that was calibrated on the rearrangement case where the lowest moving trap first reached 0.9.

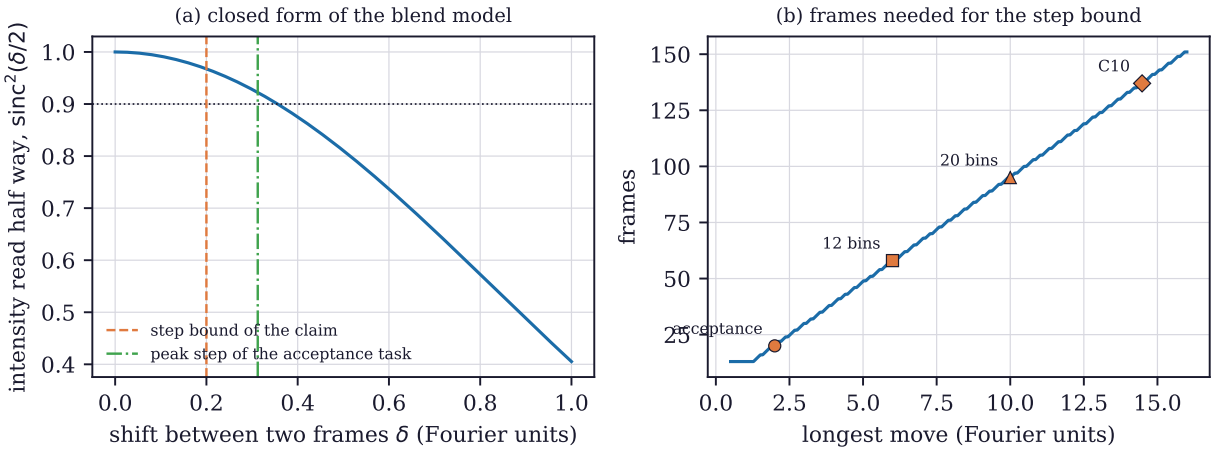


Figure 1: Theory of the transient intensity. (a) Closed form $\mathrm{sinc}^2(\delta/2)$ of the intensity of a single trap read half way between two frames shifted by δ Fourier units, with the step bound of the claim and the peak step of the acceptance task. (b) Frames that the bound requires against the length of the move, with the moves of the experiments marked.

Proposition 6. *For the minimal-jerk profile the largest shift between consecutive frames of a trap that travels L Fourier units in S intervals is $\delta_{\max} \approx \frac{15}{8} L/S$, and a step bound δ_{claim} requires $S \geq \frac{15}{8} L/\delta_{\mathrm{claim}}$ intervals; the number of frames of a sequence, and so its cost, grows linearly with the length of the move.*

Proof. The profile of Eq. 15 has $h'(\tau) = 30\tau^2(1-\tau)^2$, which is maximal at $\tau = \frac{1}{2}$ with value $\frac{30}{16} = \frac{15}{8}$. For large S the shift between frames s and $s+1$ is $L(h(\frac{s+1}{S}) - h(\frac{s}{S})) \approx Lh'(\frac{s}{S})/S \leq \frac{15}{8} L/S$. Solving $\frac{15}{8} L/S \leq \delta_{\mathrm{claim}}$ for S gives the bound. □

For the moves of the paper, Eq. 21 gives 20 frames for the acceptance move, 58 and 95 frames for the two lattice moves and 137 frames for the longest move of the rearrangement case (Fig. 1b), which are the frame counts used in Section 7.

Figure 1 shows both statements. Panel (a) shows that the closed form falls slowly at first, so a small shift costs little and the bound on the step is what keeps the intensity above 0.9; it does not show the effect of the other traps, which is measured, not derived. Panel (b) shows that the frame count rises linearly with the move, so the longest rearrangement move needs several times the frames that the bound names for the acceptance move, which is the practical limit of the method discussed in Section 9.

5.4 Phase locking and the phase path

The lock turns the free optical phases into a prescribed path. Eq. 19 shows that with a bounded phase step the target of the optimization changes slowly between frames, which a free phase does not guarantee: the fixed point of the weighted iteration determines the trap phases only through the interference of the pixels, and nothing prevents them from jumping between consecutive frames when the traps move. The ablation measures the consequence (Section 8). The lock has a price: the path of the phases is prescribed by the end phases, and a path along which no hologram can make the traps equal cannot be repaired by the per-frame optimization.

Conjecture 2. *When the flicker-bounded sequence misses the bound on a long move with the frames of Eq. 21, the cause is the phase path that the end phases prescribe, not the number of frames or the number of iterations per frame.*

The evidence is that the seeds that miss the bound on the lattice moves stay below it when the frames are doubled (the lists of Section 7), and that on the rearrangement case a seed collapses to 0.155 of the mean with the frames of the move while the others reach 0.902 to 0.917. The conjecture would be refuted by a seed that misses the bound with the frames of the move and reaches it with more frames or iterations; the data of this paper contain none. The conditions under which the end phases from phase-fixed WGS give a good path are not characterized.

Table 4: Hypotheses with the condition that refutes each, stated before the experiments. The thresholds are those of the design records of the methods and of the acceptance criteria of the engine.

ID	Statement	Refuted if
H1	On the crosstalk device, device-aware WGS lowers the spread of a 10×10 lattice at least two times against phase-fixed WGS, at an efficiency at most ten percent lower	the factor is below two or the efficiency ratio below 0.9 on a seed
H2	The gain disappears when the device model is removed from the loop	the control keeps a factor of two or more, or more iterations of the baseline close the gap
H3	On the benchmark's device, device-aware WGS meets both thresholds of H1 on the square lattice and on the large array	either geometry fails either threshold
H4	On the acceptance task, the flicker-bounded sequence keeps the lowest moving trap at 0.9 of its mean or more, above linear phase interpolation, with a per-frame spread no worse	the ratio is below 0.9 or not above that of the baseline on a seed, or the per-frame spread is larger
H5	With the frames of Eq. 21, the flicker-bounded sequence keeps the lowest moving trap at 0.9 or more on longer moves of the lattice	a majority of seeds below 0.9
H6	Removing the weights raises the spread of device-aware WGS	the spread without the weights is not higher
H7	Removing the trajectory lock brings the lowest moving trap of the flicker-bounded sequence below 0.9	the variant stays at 0.9 or more on a majority of seeds
H8	Removing the per-frame optimization brings the lowest moving trap of the flicker-bounded sequence below 0.9	the variant stays at 0.9 or more on a majority of seeds

6 Experimental design

6.1 Hypotheses

The hypotheses are stated here, before the results, with the condition that refutes each. The numbers in the statements are the thresholds of the design records of the two methods, of the research plan and of the acceptance tests of the engine. The thresholds of H1 to H4 and H6 predate the experiments reported here. The decision rules of H5, H7 and H8 (a majority of seeds) are not recorded in those documents before the runs, and the module docstring already records the lattice outcome measured on 2026-10-29 (a test run), so these three rules are treated as post hoc. Table 4 lists them. H1 is the hypothesis of the method: the spread on the crosstalk device is at least two times below that of phase-fixed WGS at an efficiency at most ten percent lower. H2 is the control that excludes the other explanation of H1, that more iterations or a different optimizer give the gain without the device model. H3 is the fuller acceptance criterion of the research plan of the method, applied to the geometries and the device of the studio's benchmark: the factor of at least two at an efficiency ratio of at least 0.9 on the square lattice and on the large array. It is stricter than H1, which concerns one lattice behind a crosstalk-only device, and it uses a harsher device, with quantization, a fill factor and flicker; the part of the criterion that concerns camera feedback and the time per iteration is not tested here. H4 is the hypothesis of the second method on the acceptance task. H5 is the transfer of the step bound from the case where it was calibrated to moves that did not take part in the calibration. H6 to H8 are the rationales of the components of the methods, stated in the design records: the weights supply an integral action that few steps lack, and both the lock and the per-frame optimization are necessary.

6.2 Data and provenance

All data are generated by the engine from seeds; no measured hologram or camera image enters the paper. The tasks are the following. The lattice task is a 10×10 square lattice of equal traps eight FFT bins apart on an SLM of 256^2 pixels padded by 2, at 850 nm, with a lens of 250 mm, a pitch of 12.5 μm and a numerical aperture of 0.5 in air, and a Gaussian input beam; it is scored on the crosstalk device (5 seeds) and on the full device (3 seeds). The benchmark geometries are those of the studio's benchmark file: a square lattice, a

ring of 36 traps, 100 random traps, a quasi-periodic set of 119 traps and a large set of a 32×32 lattice and a disordered set of 819 traps, on the device of case C08; the benchmark's runs for device-aware WGS hold 2 seeds per cell and those for phase-fixed WGS 20 (5 on G6), within a wall-clock bound per group, so the intervals on those cells are interquartile ranges of two seeds and are read as ranges. The smaller grid of the studio's case C08 is an 8×8 array on a 128^2-pixel SLM; its baked scores come from the committed manifest and the committed ranges over 3 seeds. The sequence tasks are a 4×4 lattice eight bins apart in which the two right-hand columns, eight traps, move four bins (the acceptance task, 5 seeds, 13 frames), the same lattice with the two columns moved 12 and 20 bins on 64^2 pixels (8 seeds), and the rearrangement case C10 of the studio, with 36 target sites of which 16 are reached by a move on the assignment of the Hungarian method (6 seeds). The assignment of C10 is that of the rearrangement method of the studio [27], shared by the flicker-bounded sequence and the baseline so that the two methods move the same traps along the same paths.

6.3 Methods compared

The published methods are run by the product on the same data, not quoted. For device-aware WGS the baselines are plain GS, weighted GS and phase-fixed WGS, the last at 30 and at 100 iterations, scored on the device; the reference of the comparison is the better of the two phase-fixed runs on the spread and on the efficiency (Eq. 5), because more iterations cannot remove a crosstalk that the model does not contain (Proposition 1). The ladder methods of the studio are the implementations of the cited algorithms and are checked against the independent library slmsuite [7] in the engine's tests. For the flicker-bounded sequence the baseline is linear phase interpolation as described in its source [26]: two independent phase-fixed WGS holograms of K_0 iterations, the position, amplitude and optical phase of each trap read from their fields, interpolated linearly over the same frames (the phase by its wrapped difference), and the transform inverted at every frame by the superposition of the interpolated spots without optimization. It is our implementation of a published description, not the authors' code. The controls are device-aware WGS with the device removed from the loop and the ablated variants of Section 8.

6.4 Metrics and statistics

The metrics are the efficiency and the spread of Eq. 4, the factor and the efficiency ratio of Eq. 5, the transient ratio of Eq. 6 and the per-frame spread, the largest of the relative standard deviations of the trap intensities over the frames of a sequence. The factor of a seed is a ratio, so its mean and interval are taken on the logarithmic scale: for n seeds with factors F_i,

$$\bar{F} = \exp\Big(\frac{1}{n}\sum_i \log F_i\Big), \qquad \bar{F}\, e^{\pm t_{n-1} s/\sqrt{n}}, \quad s^2 = \frac{1}{n-1}\sum_i \big(\log F_i - \overline{\log F}\big)^2, \tag{24}$$

with t_{n-1} the two-sided 95 percent quantile of Student's t distribution. The transient ratio of a method is summarized by the mean, the sample standard deviation and the interval of the mean, $\bar{\rho} \pm t_{n-1} s/\sqrt{n}$, and the comparison of the flicker-bounded sequence with the baseline is a paired difference,

$$\Delta_i = \rho_i^{\mathrm{FB}} - \rho_i^{\mathrm{LPI}}, \qquad \bar{\Delta} \pm t_{n-1}\,\frac{s_\Delta}{\sqrt{n}}, \tag{25}$$

over the seeds, where the pairing is by seed index only: the flicker-bounded sequence draws its random start from the seed, and the endpoints of the baseline are drawn from an offset seed, so the two methods do not share a random start and the paired interval is not a matched-start comparison. With five seeds the intervals are wide and the tests that they imply are of low power; no p-value is reported, and a claim rests on the interval and on the count of seeds that meet a threshold. The medians and ranges are reported as well, because the distributions are not normal (the factor of device-aware WGS is bounded below by zero and skewed). The benchmark cells report the median and the interquartile range of the studio's protocol.

6.5 Protocol and ablations

All methods are deterministic given a seed through the portable generator of the engine, so a rerun reproduces the numbers up to the floating point of the machine's FFT. The seeds are $1, \ldots, n$ with n stated per experiment: 5 for the crosstalk device, 3 for the full device, 5 for the acceptance task, 8 for the long lattice moves, 6 for the rearrangement case, and 5, 3 and 5 for the ablation of device-aware WGS on the

Table 5: Device-aware WGS on the 10×10 lattice task, scored on the device. Medians over 5 seeds for the crosstalk device and 3 for the full device; σ is the relative standard deviation of the trap intensities in percent and e the efficiency. Rows marked n/a have no entry for that device or metric.

Method	σ on the crosstalk device	σ on the full device	e (crosstalk device)
GS, 30 iterations	18.7	n/a	n/a
Weighted GS, 30 iterations	5.2	n/a	n/a
Phase-fixed WGS, 30 iterations	5.21	5.75	0.748
Phase-fixed WGS, 100 iterations	5.21	n/a	n/a
Device-aware, device model removed	5.14	5.77	n/a
Device-aware WGS	0.205	0.72	0.745

crosstalk device, of device-aware WGS on the full device and of the flicker-bounded sequence. The budget of device-aware WGS is 4 outer iterations after 12 warm iterations; that of the flicker-bounded sequence is 13 frames of 10 iterations after a cold start of 30. The ablation is planned for the components that the design records name as the rationale of each method (H6 to H8): the weight update, the device-aware phase step, the safeguard and the number of steps for device-aware WGS; the trajectory lock, the per-frame optimization, the number of iterations per frame and the time profile for the flicker-bounded sequence. Each variant removes one component and keeps the others, on the same seeds and tasks as the main result, and the variant with all components on is checked against the engine's module.

7 Results

7.1 Device-aware weighted GS on the lattice task

Table 5 and Fig. 2 give the main comparison. On the crosstalk device, phase-fixed WGS leaves a spread of 5.21 percent at 30 iterations and 5.21 percent at 100, although it reaches 0.015 percent on the ideal model: the crosstalk costs a factor of about 340, as Proposition 1 anticipates, and the baseline at 100 iterations is no better than at 30, which excludes more iterations as the explanation of any gain. Device-aware WGS reduces the spread to a median of 0.205 percent (mean 0.213 with a standard deviation of 0.052 over the seeds, largest value 0.298 percent). Against the better of the two baseline runs the factor of Eq. 5 has a geometric mean of 24.4 with a 95 percent interval of 17.5 to 34.1, a median of 26 and a range of 16 to 33, and the efficiency ratio has a median of 0.995 with a smallest value of 0.991. H1 therefore holds on every seed with a wide margin on the factor and a narrow one on the efficiency: the loss of efficiency is about half a percent against the allowed ten. The factor varies from seed to seed with the baseline's own spread, so the distribution and not one seed is the result (Fig. 2b).

The control that keeps the structure of device-aware WGS and removes the device model from the loop stays at 5.14 percent, the level of the baseline, so H2 holds: the gain comes from the device model and not from the extra iterations or from the gradient steps. Device-aware WGS pays for this on the ideal model: scored without the device its spread is 5.34 percent, because its hologram is built to cancel the crosstalk and not to be uniform before it. Figure 3 shows the same result as a trade-off: the baselines and the control sit together at a spread of several percent, device-aware WGS sits two orders of magnitude lower at the same efficiency, and no point of device-aware WGS buys its uniformity with light. The curve of Fig. 4b shows how device-aware WGS gets there on seed 1: the spread falls during the outer iterations and the best iterate is the last or one of its tail.

7.2 Full device and the small grid of the studio

With a 4-bit quantizer and a fill factor of 0.85 added, the baseline spread is 5.75 percent and device-aware WGS reaches 0.72 percent (mean 0.828, standard deviation 0.297, over 3 seeds). The geometric-mean factor is 7.0 with a 95 percent interval of 3.4 to 14.3 (median 7.9, range 5.0 to 8.6) at 0.990 of the efficiency, and the model-free control stays at 5.77 percent. The interval is wider than on the crosstalk device because the quantizer is differentiated as the identity and the three seeds are few; the factor is still above two on every seed, so H1 holds on this device as well.

The small 8×8 array on a 128^2-pixel SLM, with 4-bit phase, crosstalk, a fill factor of 0.85, 0.05 rad of flicker and a fixed budget of 30 iterations, is the device-limited case of the studio. Its committed scores are 7.04

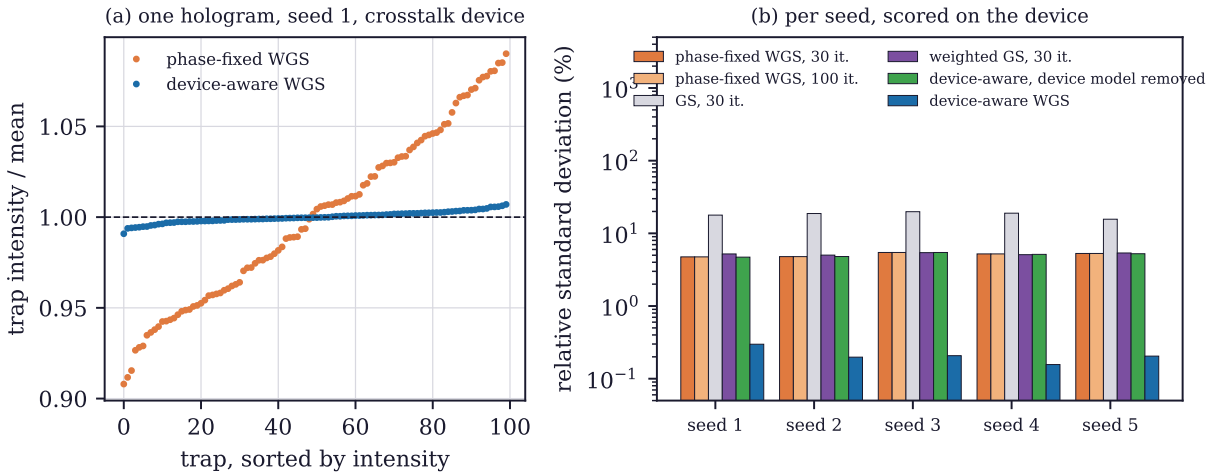


Figure 2: Device-aware WGS on the crosstalk device. (a) Sorted trap intensities of one hologram (seed 1): phase-fixed WGS leaves a spread, device-aware WGS is flat. (b) Relative standard deviation on the device for every seed on a logarithmic scale: the baselines, device-aware WGS with the device model removed from the loop, and device-aware WGS.

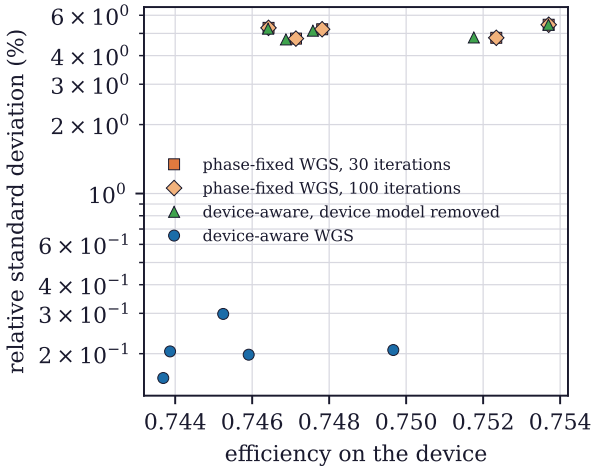


Figure 3: Efficiency against relative standard deviation on the crosstalk device, one marker per seed: phase-fixed WGS at 30 and 100 iterations, device-aware WGS with the device model removed from the loop, and device-aware WGS. The vertical axis is logarithmic.

percent for device-aware WGS and 13.42 percent for phase-fixed WGS on the bake's seed, a factor of 1.9 at an efficiency ratio of 0.966; over the 3 seeds of the committed ranges the factors are 1.906, 1.997 and 2.228, of which 1 is two or more, and the ratio of the means is 2.0. The factor is not a constant: it depends on the grid, the device, the budget and the seed. The benefit shrinks as the device departs from what the optimizer models, since flicker is scored and not modelled and the quantizer is only straight-through.

7.3 Device-aware WGS on the geometries of the benchmark

Table 6 and Fig. 5 repeat the comparison on the geometries of the studio's benchmark, on the device with flicker, at 100 iterations with the default parameters. Over the 5 geometries where both methods ran, the median factor is 1.73 (range 0.97 to 4.69) at a median efficiency ratio of 0.996, and 2 of them meet both thresholds: the ring and the random set. On the square lattice and the quasi-periodic set device-aware WGS keeps its start, with a factor near one (the list of factors is G1 0.97, G2 2.28, G3 4.69, G4 1.05 and G6 1.73), because on those geometries every iterate beyond the start gives up more than the allowed share of the light, so the safeguard is expected to return the start; the factors near one are consistent with that, but the iteration that the safeguard returned is not recorded in the table, so it is an inference and not a measurement. On the large set it lowers the spread by less than the factor of two. H3 therefore does not

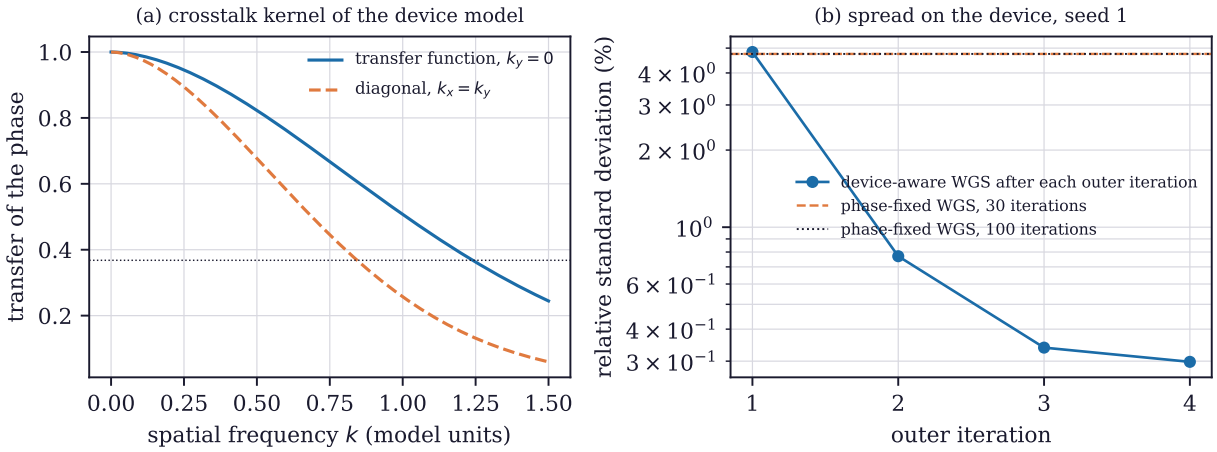


Figure 4: Device model and convergence of device-aware WGS. (a) Transfer function of the crosstalk kernel of Eq. 2 along one axis and along the diagonal, in the model's frequency unit, which is not verified against the source of the fit. (b) Relative standard deviation of device-aware WGS after each outer iteration on seed 1 against the two baseline levels.

Table 6: Device-aware WGS against phase-fixed WGS on the geometries of the studio's benchmark, on the device with flicker, at the stated iteration budget with default parameters. The spreads are medians over the seeds of each cell with the interquartile range of the cell in parentheses; device-aware WGS cells hold two seeds. The factor is the ratio of the median spreads, the efficiency ratio the ratio of the median efficiencies, and the last column states whether both thresholds of H1 are met.

Geometry	Traps	Seeds (device-aware, phase-fixed)	σ phase-fixed WGS (%)	σ device-aware WGS (%)	Factor	e ratio	Both thresholds
G1 Square 10 x 10	100	2, 20	26.44 (25.09 to 27.55)	27.30 (26.34 to 28.26)	0.97	1.013	no
G2 Ring of 36 traps	36	2, 20	16.96 (16.12 to 17.94)	7.45 (7.25 to 7.65)	2.28	0.965	yes
G3 Random 100 traps	100	2, 20	20.67 (20.24 to 20.83)	4.41 (4.38 to 4.44)	4.69	0.938	yes
G4 Quasi-periodic 119	119	2, 20	85.16 (63.96 to 117.28)	81.09 (73.26 to 88.92)	1.05	0.996	no
G6 Large: 32 x 32 and disordered 819	1024	2, 5	29.65 (9.16 to 29.73)	17.11 (10.73 to 23.50)	1.73	1.061	no

hold: the square lattice and the large array fail the factor. The failure is the opposite of what the lattice task of the previous subsections would suggest, and it is the main limit of the method.

Two reasons keep the reading cautious. The device-aware WGS cells hold two seeds, so the interquartile ranges in the table are ranges of two values and the factor of a geometry is a ratio of medians with no interval. And the device of the benchmark differs from the one of H1 in the quantizer, the fill factor and the flicker, so a failure on the square lattice there is not a contradiction of H1 on the crosstalk device; it is the measurement of the scope.

7.4 Flicker-bounded sequences on the acceptance task

With 13 frames the lowest transient ratio of the moving traps is a mean of 0.922 (standard deviation 0.007, interval 0.913 to 0.931, median 0.920, range 0.915 to 0.932) for the flicker-bounded sequence over 5 seeds, above 0.9 for every seed, against a mean of 0.644 (standard deviation 0.063, interval 0.565 to 0.723, range 0.551 to 0.709) for linear phase interpolation, which is below 0.9 for every seed. The paired difference of Eq. 25 has a mean of 0.278 with a 95 percent interval of 0.198 to 0.358, so the interval excludes zero. The per-frame uniformity differs by a factor of 50 to 70: the largest per-frame relative standard deviation is at most 0.39 percent for the flicker-bounded sequence (median 0.34) and between 19 and 27 percent for linear phase interpolation (median 22.8), because the intermediate frames of linear phase interpolation are not optimized. H4 therefore holds on the acceptance task. Figure 6a shows the trajectory of the worst trap, which stays near its mean for the flicker-bounded sequence and dips for the baseline; it does not show the other traps. The measured mean of the flicker-bounded sequence matches the single-trap closed form of Eq. 23 at the peak step of this task, 0.922, which suggests that at this step the interference of the other traps adds little.

The bound depends on the number of frames (Fig. 6b). With seven frames the bound of 0.9 is not met: the medians are 0.746 for the flicker-bounded sequence (smallest 0.735) and 0.643 for linear phase interpolation (largest 0.666), so the flicker-bounded sequence stays above the baseline and the hypothesis as stated is

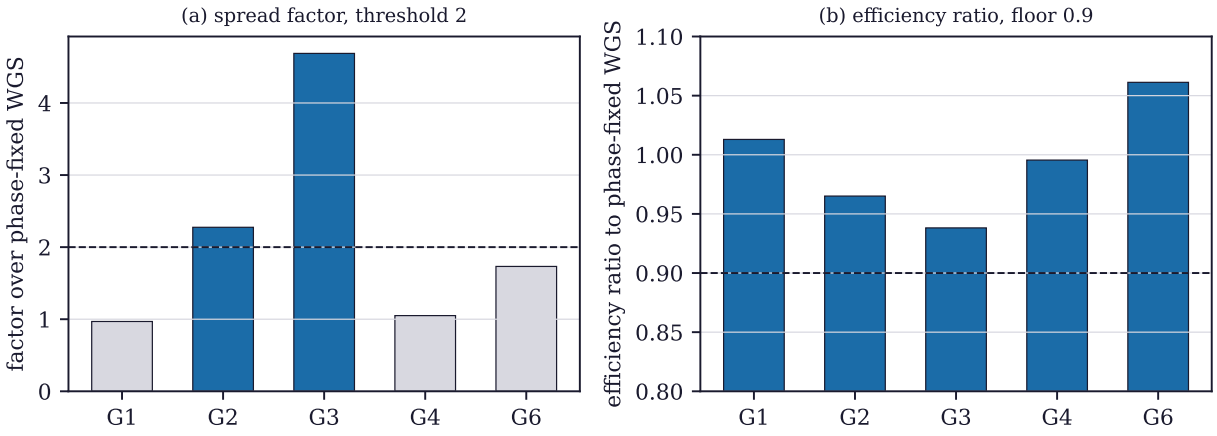


Figure 5: Device-aware WGS on the benchmark geometries of the device with flicker. (a) Factor of the spread over phase-fixed WGS with the threshold of two. (b) Ratio of the efficiencies with the floor of 0.9. Filled bars meet the threshold; the benchmark has no geometry G5, which is three-dimensional and outside the scope of device-aware WGS.

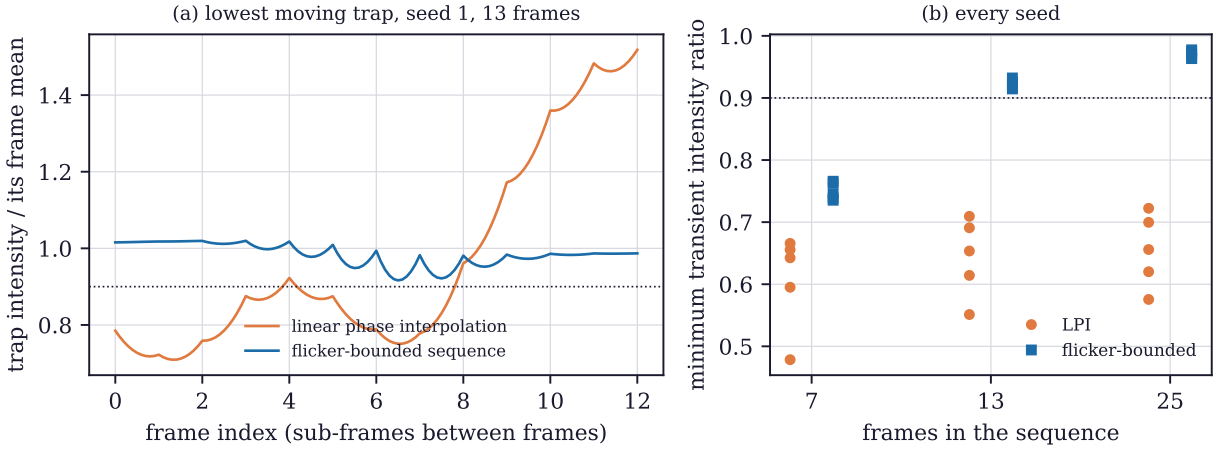


Figure 6: Flicker-bounded sequence on the acceptance task. (a) Intensity of the moving trap with the lowest minimum, divided by its frame mean, along the 13-frame sequence (seed 1): linear phase interpolation dips and overshoots, the flicker-bounded sequence stays close to one; the dotted line is 0.9. (b) Minimum transient ratio of the moving traps against the length of the sequence, one marker per seed.

refuted at that length, where each frame moves the traps farther. With twenty-five frames the medians are 0.970 for the flicker-bounded sequence and 0.656 for linear phase interpolation, and the smallest value of the flicker-bounded sequence is 0.963 against a largest value of 0.722 for the baseline. The baseline does not improve with more frames, because its frames are interpolations of two fixed endpoints and not optimized holograms; the flicker-bounded sequence improves with every added frame, until the closed form of Eq. 23 approaches one.

7.5 Long moves and the rearrangement case

The acceptance task moves the traps four bins in 13 frames, a peak step of 0.31 Fourier units per frame, whereas the step bound of the claim is 0.2 and would name 20 frames. The scope of the claim is measured on longer moves, with the end phases taken from phase-fixed WGS at the destination and the bound on the phase step per frame lifted from 0.3 to 1 rad for every run, so that runs at the default frames, whose step the default bound refuses on the longer move, can be compared (at the frames of the move the default bound holds). On the same lattice with two columns moved 12 and 20 bins (6 and 10 Fourier units), the 13 default frames leave the lowest moving trap at most at 0.528 over 8 seeds, so the default frames do not suffice. With the 58 and 95 frames of the move (peak steps of 0.197 and 0.199 Fourier units) the medians are 0.933 and 0.936, and 6 and 6 of the 8 seeds reach 0.9 (Table 7, Fig. 7a,b). H5 holds on a majority of

Table 7: The flicker-bounded sequence and linear phase interpolation on long moves: the lattice with two columns moved 12 and 20 bins, and the rearrangement case C10 at an equal number of frames and at the frames of the move. The frame counts are those of Eq. 21 except the equal-frames row; the minimum ratio is that of the lowest moving trap per seed, over eight seeds for the lattice and six for C10, and the last column counts the seeds where the flicker-bounded sequence is above linear phase interpolation.

Move	Frames	Peak step (Fourier units)	Sequence median	Sequence range	Seeds at 0.9 or above	Linear phase interpolation range	Sequence above it
lattice, 12 bins	58	0.197	0.933	0.023 to 0.965	6 of 8	0.568 to 0.756	7 of 8
lattice, 20 bins	95	0.199	0.936	0.584 to 0.966	6 of 8	0.581 to 0.701	7 of 8
rearrangement C10, 16 moving traps	25	1.131	0.335	0.205 to 0.391	0 of 6	0.320 to 0.594	0 of 6
rearrangement C10, 16 moving traps	137	0.200	0.910	0.155 to 0.917	5 of 6	0.360 to 0.690	5 of 6

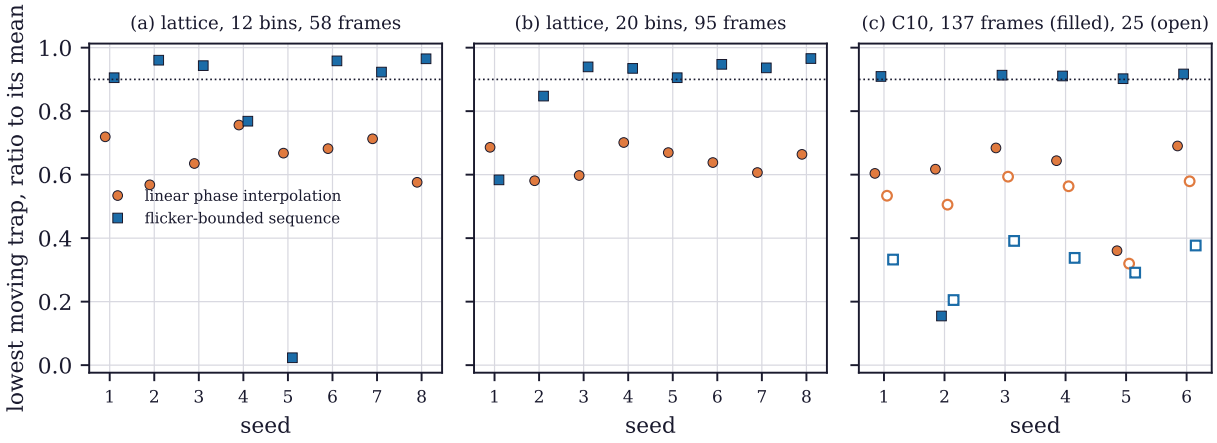


Figure 7: Lowest moving trap of the flicker-bounded sequence and of linear phase interpolation per seed on long moves. (a) and (b) Lattice with two columns moved 12 and 20 bins, with the frames of the move. (c) Rearrangement case C10 with the frames of the move (filled markers) and with 25 frames (open markers). The dotted line is 0.9.

seeds. The seeds that miss the bound (0.768 and 0.023 at 12 bins; 0.584 and 0.847 at 20 bins) miss it on the phase path that their end phases prescribe: with twice the frames they reach 0.757 and 0.000 at 12 bins; 0.558 and 0.840 at 20 bins, so the frame count is not the cause (Conjecture 2). Linear phase interpolation on the same moves, assignment and frames reaches 0.568 to 0.756 and 0.581 to 0.701, and the flicker-bounded sequence is above it on 7 and 7 of the 8 seeds.

On the rearrangement case of the studio (C10: 36 target sites, 16 of them reached by a move on the assignment of the Hungarian method, the longest 14.5 Fourier units), held phases dim the moving traps to 0.177 at 25 frames on the first seed. With the end phases at 25 frames, a peak step of 1.13 Fourier units, linear phase interpolation on the same assignment leads on 6 of 6 seeds (the flicker-bounded sequence at most 0.391, the baseline 0.320 to 0.594). With the 137 frames of the move (a peak step of 0.200 Fourier units), the flicker-bounded sequence reaches 0.902 to 0.917 (median 0.911) on 5 of the 6 seeds and leads linear phase interpolation (0.360 to 0.690) on 5; the seed that misses is seed 2 at 0.155 against 0.617 for LPI. The step bound was calibrated on this case, so the result on C10 is not an independent test; the lattice moves are, and they support the bound on a majority of seeds and not on all. At an equal number of frames on a long move linear phase interpolation leads, which Fig. 7c shows seed by seed.

7.6 Outcome of each hypothesis

Table 8 summarizes the outcome of the hypotheses of Section 6 with the evidence for each. Of the eight hypotheses, five hold in the form stated (H1, H2, H4, H5, H8) and three are refuted by their own conditions (H3, H6, H7); H6 and H7 are measured in the ablation of Section 8. The refutations are the findings with the most consequence: device-aware WGS does not meet its thresholds on the square lattice and the large array of the benchmark's device, the weights that the design of device-aware WGS keeps are not what produces its gain, and the lock is not necessary on a majority of the seeds of the acceptance task, although it removes the worst of them. The holding hypotheses are the ones that the design records stated for the methods on their own tasks: the factor of device-aware WGS on the lattice behind a crosstalk device with its control, and the transient ratio of the flicker-bounded sequence on its acceptance task, both with the intervals reported above. Every refutation concerns a scope or a rationale: a geometry, a device, or a component of the method.

Table 8: Outcome of the hypotheses of Table 4 with the evidence from the experiments. The numbers are those of Tables 5 to 7 and Section 8.

ID	Outcome	Evidence
H1	holds	factor 16 to 33 on 5 seeds, efficiency ratio at least 0.991; 5.0 to 8.6 on the full device
H2	holds	control at 5.14 percent against 5.21 percent for the baseline
H3	refuted	factors G1 0.97, G2 2.28, G3 4.69, G4 1.05 and G6 1.73 on the benchmark device
H4	holds	ratio 0.915 to 0.932 against 0.551 to 0.709; frame spread at most 0.39 against 19 to 27 percent
H5	holds on a majority	6 and 6 of 8 seeds at 0.9 or more with the frames of the move
H6	refuted	spread 0.030 percent without the weights against 0.205 with them
H7	refuted	without the lock the lowest moving trap is at 0.9 or more on three of five seeds (mean 0.877, worst 0.776)
H8	holds	without per-frame optimization 0.786 on average, below 0.9 on every seed

Table 9: Ablation of device-aware WGS on the crosstalk device, 10×10 lattice task, 5 seeds. Each row removes one component or changes one parameter; σ is the relative standard deviation on the device, the factor is the geometric mean over the better of the two phase-fixed baseline runs with its 95 percent interval, and the last column is the median ratio of the efficiencies.

Variant	σ median (%)	σ range (%)	Factor over phase-fixed WGS, 95% interval	e ratio
device-aware WGS, all components	0.205	0.157 to 0.298	24.4 (17.5 to 34.1)	0.995
without the weight update	0.030	0.016 to 0.032	201.3 (132.9 to 304.9)	0.999
without the device-aware phase step	2.771	2.245 to 2.970	1.9 (1.6 to 2.3)	1.007
without the best-iterate safeguard	0.205	0.157 to 0.298	24.4 (17.5 to 34.1)	0.995
device model removed from the loop	5.137	4.706 to 5.454	1.0 (1.0 to 1.0)	1.000
1 conjugate-gradient step per iteration	1.741	1.121 to 2.313	3.0 (2.1 to 4.2)	0.996
6 conjugate-gradient steps per iteration	0.034	0.016 to 0.102	139.8 (60.7 to 322.0)	0.996
warm start only (no iteration)	6.508	4.861 to 7.418	0.8 (0.7 to 1.0)	1.000

8 Ablation

8.1 Components of device-aware weighted GS

The ablation removes one component of device-aware WGS at a time on the lattice task and the seeds of the main result. All variants share the start, the budget and the scoring, and the variant with all components on reproduces the module of the engine to a difference of 0.0e+00 on the spread for every seed, so the rows are the module with components removed. Table 9 gives the crosstalk device and Table 10 the full device; Fig. 8a shows the per-seed spreads. The statistics are the median and the range of the spread, the geometric-mean factor over the reference of Eq. 5 with its 95 percent interval, and the median efficiency ratio, for 5 seeds on the crosstalk device and 3 on the full device.

The device-aware phase step carries the gain. Without it, the weights read on the device alone leave a median spread of 2.771 percent, a factor of 1.9 (interval 1.6 to 2.3) over the baseline: measuring the amplitudes on the device is worth about two times, and the phase step is worth the remaining factor, which is 14 between the variant without the step and the full method. With the device model removed from the loop the spread returns to the level of the baseline, 5.137 percent, 25 times above the full method. The gain therefore comes from putting the device in the phase update; the weights alone do not provide it.

The weights do not help, they hurt. Without the weight update the median spread is 0.030 percent (range 0.016 to 0.032), 7 times lower than the 0.205 percent of the full method, at an efficiency ratio of 0.999; the geometric-mean factor over the baseline rises from 24.4 (interval 17.5 to 34.1) to 201.3 (interval 132.9 to 304.9). On the full device the spread without the weights is 0.242 percent against 0.724 percent with them, 3 times lower. H6 is therefore refuted on both devices: the integral action of the weights, which the design of the method gives as its reason, is not needed when three conjugate-gradient steps follow the target on this task, and it costs spread. This is in line with Conjecture 1, since six steps per outer iteration reach 0.034 percent and one step leaves 1.741 percent, so the number of steps, and not the weights, sets the quality; the conjecture itself, that the weights wind up, is compatible with these data and is not proved by them. The same comparison on the lattices of the benchmark answers whether the variant without weights would do

Table 10: Ablation of device-aware WGS on the full device (4-bit phase, crosstalk, fill factor 0.85), 3 seeds, same task and statistics as Table 9 for the components that matter; the intervals use only three seeds and are wide.

Variant	σ median (%)	σ range (%)	Factor over phase-fixed WGS, 95% interval	e ratio
device-aware WGS, all components	0.724	0.597 to 1.163	7.0 (3.4 to 14.3)	0.990
without the weight update	0.242	0.202 to 0.248	24.2 (16.8 to 34.8)	0.997
without the device-aware phase step	3.487	3.344 to 3.922	1.6 (1.3 to 1.9)	1.006
without the best-iterate safeguard	0.724	0.597 to 1.163	7.0 (3.4 to 14.3)	0.990
device model removed from the loop	5.769	5.189 to 5.781	1.0 (1.0 to 1.0)	0.999
warm start only (no iteration)	7.288	5.115 to 8.320	0.8 (0.5 to 1.3)	0.999

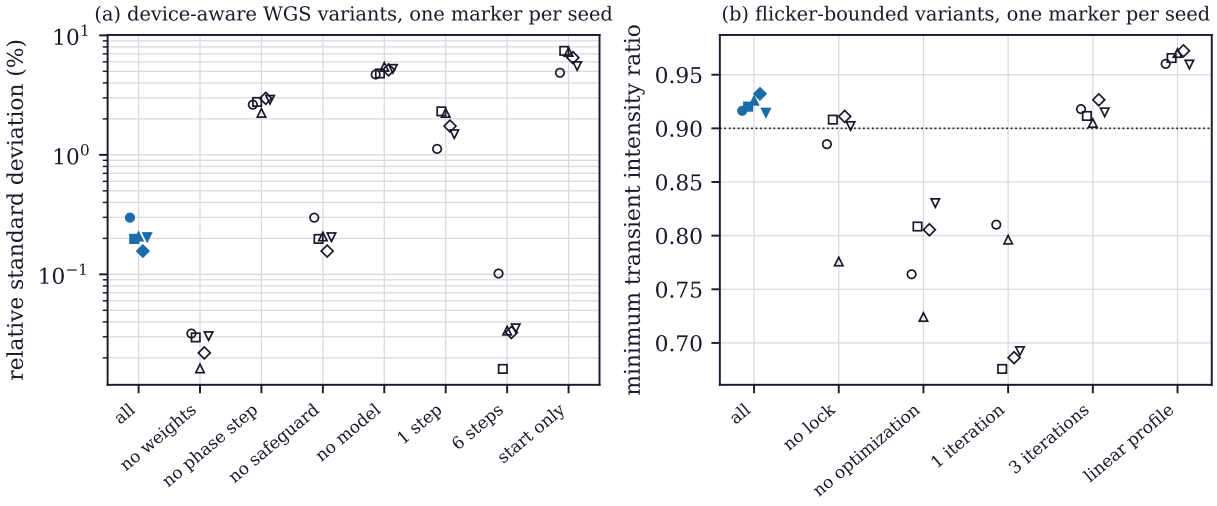


Figure 8: Ablation per seed. (a) Relative standard deviation on the crosstalk device of the variants of device-aware WGS (logarithmic axis). (b) Lowest transient ratio on the acceptance task of the variants of the flicker-bounded sequence with the line at 0.9. Filled blue markers are the full methods; one marker per seed.

better there. Each cell is the benchmark's own record, run through the engine's runner and scored against the C08 device (4-bit phase, crosstalk, fill factor and flicker) at 10 outer iterations for 3 seeds, with the weight update switched by the module's `weights` option (Table 11). On the ideal model the two forms coincide (spreads at most 0.004 percent). On the device the outcome depends on the geometry: on the square lattice the variant without weights halves the median spread, 13.64 against 29.21 percent, with the lower spread on every seed, at a lower efficiency, 0.563 against 0.598; on the ring of 36 traps the weights are better on every seed, 7.79 against 8.90 percent; on the random set the two differ by a few percent of the spread (4.43 and 4.27 percent); on the quasi-periodic set they are identical on two seeds out of three. Neither form dominates on these lattices. The evidence of this paper does not support the weight update on the acceptance task (a spread 7 times higher with it) or on the full device (3 times) and is mixed on the benchmark lattices: the variant without weights has the lower spread on the square lattice and the random set and the higher on the ring. The method is defined with the weight update and the module exposes the variant without it as an option; the weights are the first component to reconsider. Table 11 has no phase-fixed WGS reference at the same budget of 10 outer iterations, so it does not tell whether dropping the weights removes the failure of H3 on the square lattice and the large array; that question stays open.

The safeguard did not act on these seeds: with and without it the spreads are identical (0.205 percent), as Proposition 4 allows, since the last iterate is the best admissible one on the crosstalk device. Its value is on the geometries of the benchmark where the last iterate would have lost too much light. The warm start alone, 12 weighted iterations on the ideal model, leaves a median spread of 6.508 percent, a factor of 0.8 against the baseline of 30 and 100 iterations, so the device-aware iterations, not the start, produce the result.

8.2 Components of flicker-bounded sequences

Table 12 gives the ablation of the flicker-bounded sequence on the acceptance task with 5 seeds, again with all components on reproducing the module of the engine to 0.0e+00. Removing the per-frame optimization, that is, one inversion of the locked field with unit weights per frame, leaves the lowest moving trap at a mean

Table 11: Device-aware WGS with and without the weight update on the lattices of the benchmark, C08 device, 10 outer iterations: median over 3 seeds of the relative standard deviation σ and of the efficiency e on the device, and the number of seeds on which the variant without weights has the lower spread.

Geometry	σ with weights (%)	σ without (%)	e with	e without	seeds lower without
G1, Square 10 x 10	29.21	13.64	0.598	0.563	3 of 3
G2, Ring of 36 traps	7.79	8.90	0.453	0.464	0 of 3
G3, Random 100 traps	4.43	4.27	0.401	0.408	3 of 3
G4, Quasi-periodic 119	65.43	65.43	0.493	0.493	1 of 3

Table 12: Ablation of flicker-bounded sequences on the acceptance task (4×4 lattice, two columns moved four bins, 13 frames, 5 seeds). Minimum ratio is that of the lowest moving trap; the last column is the median over the seeds of the largest per-frame relative standard deviation. The last row is linear phase interpolation on the same seeds.

Variant	Minimum ratio, mean ± s.d.	Range	Seeds at 0.9 or above	Largest frame σ (%)
flicker-bounded sequence, all components	0.922 ± 0.007	0.915 to 0.932	5 of 5	0.34
without the trajectory lock	0.877 ± 0.057	0.776 to 0.911	3 of 5	2.39
without the per-frame optimization	0.786 ± 0.042	0.724 to 0.830	0 of 5	18.95
1 iteration per frame	0.732 ± 0.065	0.676 to 0.810	0 of 5	18.65
3 iterations per frame	0.915 ± 0.008	0.905 to 0.927	5 of 5	3.72
linear time profile for the positions	0.966 ± 0.006	0.959 to 0.972	5 of 5	0.26
linear phase interpolation (reference)	0.644 ± 0.063	0.551 to 0.709	0 of 5	22.8

of 0.786 (range 0.724 to 0.830), below 0.9 on every seed, and the largest per-frame spread of 18.95 percent is that of an unoptimized sequence, close to the baseline; H8 holds. One iteration per frame is not enough either (mean 0.732), and three iterations already reach a mean of 0.915 (range 0.905 to 0.927), above 0.9 on every seed, so the default of 10 iterations is more than the acceptance task needs; the per-frame spread with three iterations is 3.72 percent against 0.34 for the default, so the extra iterations buy uniformity of the frames and not transient intensity.

Removing the trajectory lock lowers the mean to 0.877 (standard deviation 0.057, range 0.776 to 0.911) and raises the per-frame spread to 2.39 percent; it stays at 0.9 or more on three of five seeds, so H7 is refuted by its own condition: the lock is not necessary on a majority of seeds, but it removes the seed-to-seed variation that a free phase brings, with a worst seed of 0.776 against 0.915 for the flicker-bounded sequence. The minimal-jerk profile of the positions costs transient intensity on this metric: a linear time profile, which moves the traps at constant speed, gives a mean of 0.966 (range 0.959 to 0.972), above the full method, because the peak step of the minimal-jerk profile is 15/8 times the mean step and the transient ratio is limited by the peak step (Proposition 6). The minimal-jerk profile is retained in the method for the smoothness of the acceleration, which the transient model does not score, and this is a choice that the ablation does not support in the metric of the paper.

9 Discussion

9.1 Why the methods behave as they do

The ideal-model solver equalizes a field that the device does not produce (Proposition 1), and the device-aware step equalizes the field that is scored. The ablation separates two contributions: the amplitudes read on the device give a factor of about two on their own, and the gradient steps through the crosstalk adjoint give the rest. The control with the device removed from the loop confirms that the steps are not the cause: the same steps on the ideal model leave the spread at the baseline. For the flicker-bounded sequence the transient ratio is limited by the shift between consecutive frames, as the closed form of Eq. 23 states, and the optimization and the lock make the frames individually uniform and mutually continuous, so that a bound on the step is a bound on the dip. The no-optimization variant fails with the same closed-form shifts as the full method, so the gain of the per-frame optimization is the uniformity of the frames, not a smaller shift.

9.2 Failure cases and negative results

Four results are negative and they bound what the paper claims. First, device-aware WGS fails the factor on the square lattice and on the large array of the benchmark's device (it keeps its start on the square lattice

and lowers the spread by a factor of 1.73 on the large array, short of two), so the success on the lattice behind a crosstalk device is specific to that device (Section 7). Second, the weights of device-aware WGS hurt on the acceptance task and on the full device, where they were supposed to help, and are mixed on the benchmark lattices (Section 8). Third, on the rearrangement case at an equal number of frames linear phase interpolation leads the flicker-bounded sequence on every seed, and the flicker-bounded sequence needs the end phases and the frames of Eq. 21 to be competitive there. Fourth, the flicker-bounded sequence collapses on a seed of the rearrangement case and on seeds of the lattice moves, and more frames do not recover them. The last result is the cleanest statement of the limit of the lock: where the end phases prescribe a bad path, the per-frame optimization cannot repair it. A method that chose the end phases jointly with the frames would remove the limit; we did not build one.

9.3 Relation to the published methods

Two published results frame what the paper adds. Schroff et al. put the crosstalk model inside conjugate-gradient minimization, lowered the error of a shaped potential, paid about 20 percent of the efficiency and ran about six times longer [21]. The variant of device-aware WGS without the weight update is the point-trap counterpart of that idea in our engine, and on the acceptance task it reaches a median spread of 0.030 percent at an efficiency ratio of 0.999 to the baseline, against 0.205 percent for the method with its weights. The two tasks, devices and metrics differ, so the numbers are not comparable, and the time per iteration was not measured here; what the comparison supports is only that the device model inside a conjugate-gradient step, and not the weighted-GS structure around it, carries the gain on this task. The phase-fixed rule of Kim et al. [4] remains the right start and the right baseline: it is the reference of every factor in this paper, and the warm start of device-aware WGS uses that rule.

For sequences the comparison is with the linear phase interpolation of Knottnerus et al. [26]. The per-frame optimization with locked phases lifts the lowest moving trap above the bound on the acceptance task under our blend model, but the source measured on hardware and our model is our own definition, so the paper establishes an ordering of the two methods under one model and not an improvement on the hardware of the source. Where the end phases prescribe a bad path, as on seeds of the rearrangement case, the proposed sequence collapses, and at an equal number of frames on that case linear phase interpolation leads; the proposed sequence is not to be used there without the frames and the end phases of Eq. 21.

9.4 What the results imply for use

The results suggest a division of labour. For a static array behind a device whose crosstalk is modelled, the device-aware phase step is the part to adopt, and the evidence of this paper favours it without the weights on the acceptance task and on the full device and is mixed on the benchmark lattices; the safeguard is cheap insurance on geometries where the last iterate loses light. For a rearrangement with moves of a few Fourier units the flicker-bounded sequence keeps the lowest trap near its mean with a number of frames that follows from Eq. 21; for long moves the number of frames, and with it the cost, grows linearly, and the choice of end phases decides the outcome seed by seed. Neither method replaces camera feedback: the paper measures hologram quality in a model, and a flat hologram on the device model is a better start for feedback, not a substitute for it. What would change these conclusions is a measured device model: with a different kernel, a different quantizer or a measured flicker the factors would change, and the sensitivity of device-aware WGS to a mismatched kernel was not tested.

10 Limitations

10.1 Threats to validity

Internal validity. The seeds are few: 5 for the crosstalk device, 3 for the full device and for the long-move statistics up to 8, so the intervals are wide and no significance test is reported. The device-aware WGS cells of the benchmark hold two seeds, so the geometry results are ranges. The step bound of the flicker-bounded sequence was calibrated on the rearrangement case and is tested out of sample only on the lattice moves. The baseline of linear phase interpolation is our implementation of a published description, scored by our own transient model, not the authors' code on their hardware; the reference value of the source (above 0.70 of the intensity on hardware) is higher than the value that the baseline scores here, so the absolute values of

the two are not comparable. The variants of the ablation are re-implementations of the loops of the modules and were verified to reproduce them, so the ablated rows are not a different method.

External validity. Every result is simulated. The crosstalk parameters come from a published fit on another device, and their frequency unit is carried over from the model and is not verified against the source; the 4-bit quantizer, the fill factor and the flicker have no published fit in this model, and random flicker is scored and not modelled by the optimizer. An instrument with another kernel would give another factor. The tasks are lattices of equal traps on bins at one wavelength and one pitch, a Gaussian beam and no aberrations; the benchmark adds other geometries on one device, and the three-dimensional and shaped-field targets are outside the scope of device-aware WGS.

Construct validity. The spread and the efficiency are model quantities, and the factor measures them against phase-fixed WGS without camera feedback, whereas the experiments close the gap with feedback; a method that is two times flatter before feedback may or may not need fewer feedback steps, and that was not measured. The transient ratio is defined by the blend model of the SLM field between frames, which omits the temporal response and the refresh of the device; it establishes an ordering of the two methods under one model and is not a prediction of atom survival. A bound of 0.9 on the transient ratio is a design threshold and is not derived from the survival of atoms.

10.2 Scope and what is not claimed

Neither method is shown to improve a camera-corrected experiment or the survival of atoms. Device-aware WGS is built for traps on FFT bins at the focal plane without vortices and costs about u^2 times a plain iteration per step, so it is baked and not live. The authors' conjugate-gradient minimization with a crosstalk model, with its automatic differentiation and its shaped potentials, is not run; the variant of device-aware WGS without the weight update is the nearest thing to it that the paper runs, on point traps, and the source reports about ten minutes per hologram for its own method. The flicker-bounded sequence is shown on lattices and one rearrangement case in two dimensions, with end phases from phase-fixed WGS, and it does not choose the assignment of atoms to sites. The time per iteration of device-aware WGS against the baseline was not measured, and the claim that device-aware WGS is within a factor of three of the baseline in time is not made. The paper also does not test the sensitivity of device-aware WGS to a kernel that differs from the one the device displays, which is the case in any real instrument whose kernel is fitted with error; a hologram optimized on a wrong kernel would carry part of the crosstalk error that the method removes in the model.

11 Reproducibility

The code, the data and the committed artifacts are in the repository `https://github.com/fsantibanezleal/CEFOP_DinHot` at release 0.02.003 (Apache-2.0); the studio runs at `https://dinhot.fasl-work.com`. Every number, table and figure of the paper is produced by two scripts from the repository root. The measurement script `manuscripts/device-aware-flicker-bounded/data/measure.py` runs the methods of the engine and writes `beyond.json`, the macro file read by the text and the result tables; with no argument it measures every part, with `-parts=ablation` it measures only the ablation and keeps the stored parts (the main measurements of the two methods took about 16 minutes in total and this ablation run about 58 minutes on the development machine, CPU only, with five seeds for device-aware WGS on the crosstalk device, three on the full device and five for the flicker-bounded sequence), and with `-macros-only` it rewrites the macros and the tables from the stored measurements. The figure script `manuscripts/device-aware-flicker-bounded/figures/make_figs.py` writes every figure as a vector PDF from `beyond.json`. The document builds with three passes of `pdflatex` from `tex/`. The benchmark file and the committed manifests that the script reads are in `data/derived/`. The acceptance tests of the two methods are `test_b1_device_wgs.py` and `test_b2_flicker_bounded.py`, and the ablated variants are checked against the modules inside the measurement script. The comparison of the weight update on the benchmark lattices (Table 11) is written by `manuscripts/device-aware-flicker-bounded/data/b1_weights_geo.py`, one process per geometry (`run G1` to `run G4`, about half an hour each) and then `write`. All methods are deterministic given a seed, so a rerun reproduces the numbers up to the floating point of the machine's FFT.

12 Conclusions

Putting the pixel-crosstalk kernel and the phase quantizer inside the phase update of weighted GS lowers the trap-intensity spread on the device by a geometric-mean factor of 24.4 (95 percent interval 17.5 to 34.1) on a 10×10 lattice behind a crosstalk device, at 0.995 of the baseline efficiency, and the gain disappears when the device model is removed from the loop. The gain does not carry to every geometry: on the benchmark's device with flicker the method meets both thresholds on the ring and the random set, keeps its start (factors near one) on the square lattice and the quasi-periodic set, and lowers the spread by a factor of 1.73 on the large set, short of the threshold of two. The ablation shows that the gradient step through the device, and not the weights, produces the gain; the variant without the weights has the lower spread on the acceptance task and the full device and a mixed one on the benchmark lattices, so the weights are the first component to reconsider. Optimizing each frame with locked, smoothly varying trap phases keeps the moving traps of the acceptance task at a mean of 0.922 of their intensity (interval 0.913 to 0.931) where linear phase interpolation falls to 0.644; with the frames that keep the peak step at most 0.2 Fourier units it meets the bound on most seeds of longer moves and not on all, and at an equal number of frames on a long rearrangement linear phase interpolation leads. The next steps are the measurement on a device with a fitted model and a camera in the loop, the ablation of device-aware WGS on the benchmark lattices with a phase-fixed WGS reference at the same budget, to test whether dropping the weights removes the failure of H3 (Table 11 has no such reference), and a choice of end phases that is joint with the frames.

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